Advanced Topics of Deep Generative Models

https://hku-data8018.github.io/

Bo Dai

Spring 2026

- Slides reference
  - Advanced Topics of Deep Generative Models, https://jiataogu.me/DeepGenerativeModels/

Impressive Conditional Diffusion Models

Text-to-image generation

DALL·E 2

"a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese"

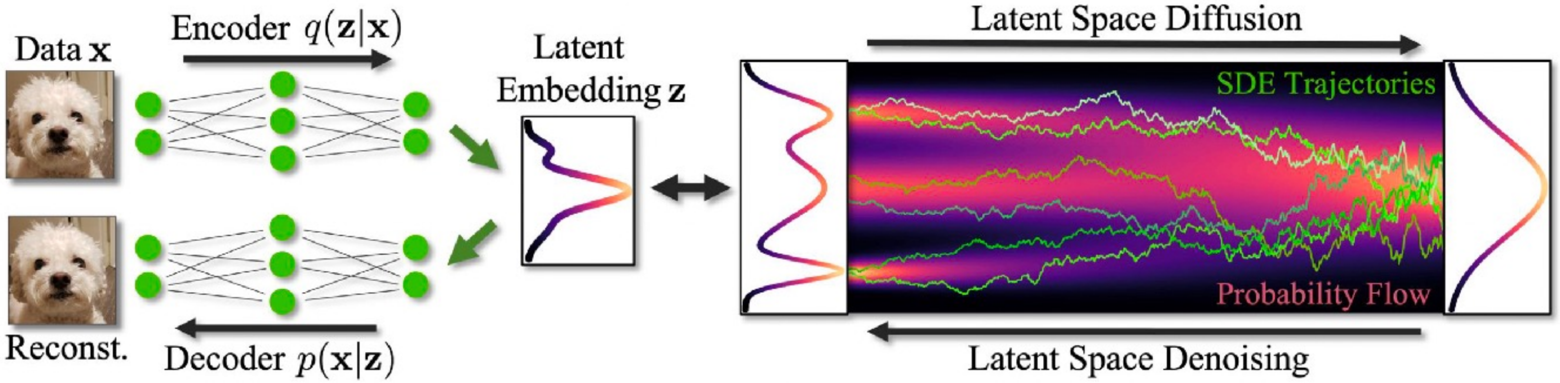
IMAGEN

"A photo of a raccoon wearing an astronaut helmet, looking out of the window at night."

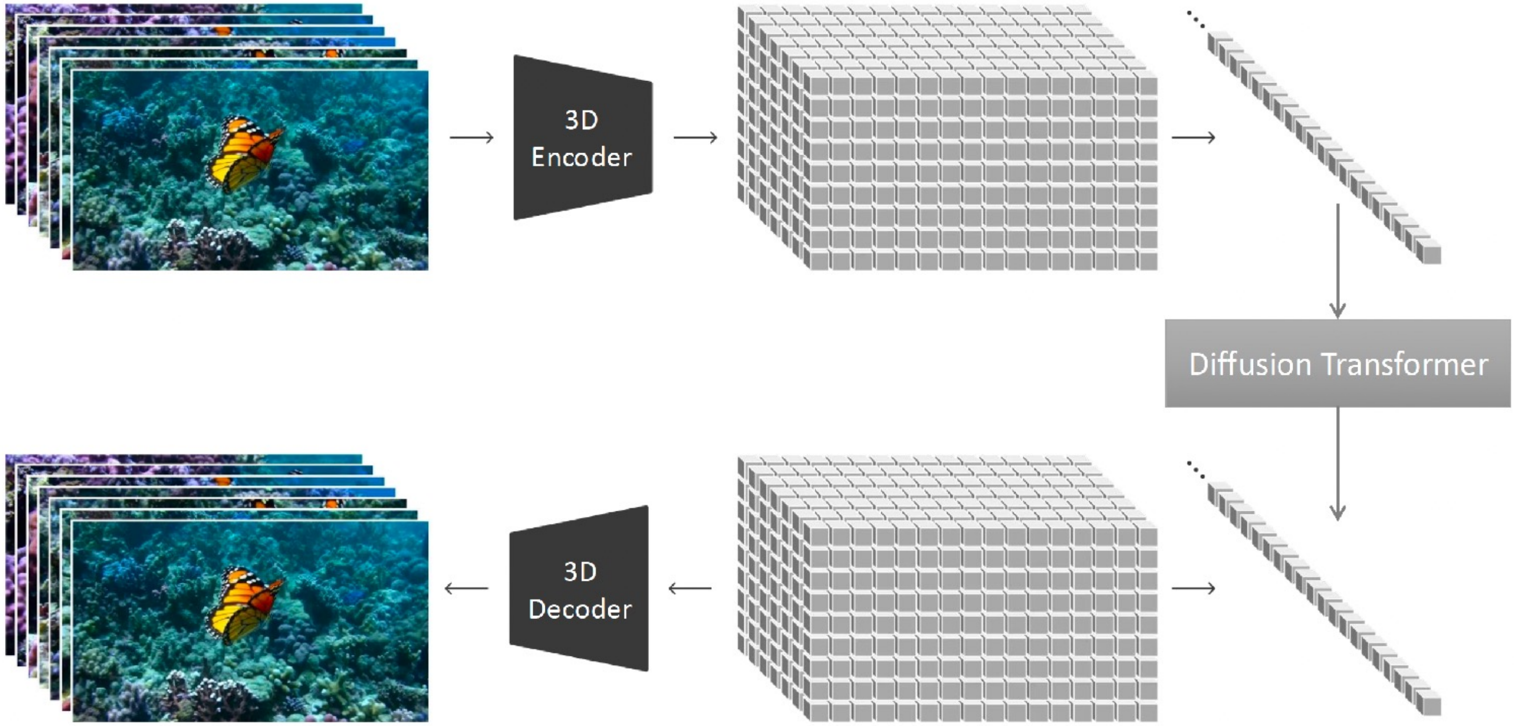
Ramesh et al., "Hierarchical Text-Conditional Image Generation with CLIP Latents", arXiv 2022.
Saharia et al., "Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding", arXiv 2022.

Two-Stage Training

- Stage I: Train a "Variational" Auto-Encoder, typically with GAN
  - GAN loss + Reconstruction loss + KL loss
  - It's a "V"AE because the KL loss has a very small β (typically < 0.01)
  - 1% "Variational", 99% Auto-encoder
- Stage II: Train a Diffusion Model in the latent space

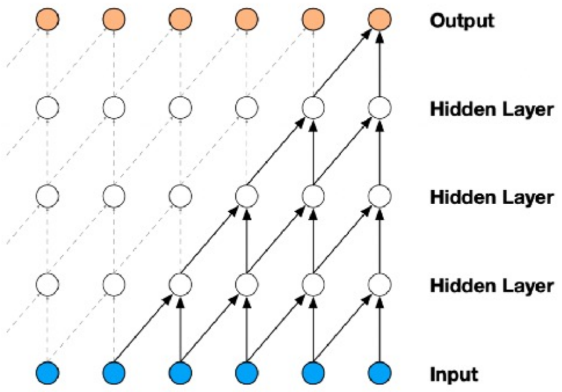


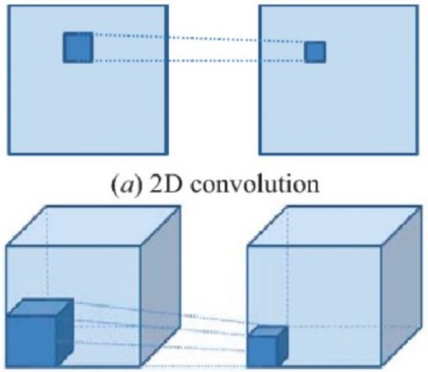
Video Diffusion Models



Temporally Causal "V"AE

- Each latent frame only depends on past video frames, but not future ones.
- Benefits:
  - Easier to encode/decode videos of various lengths.
  - Reuse image VAE/data.
- Causal 3D Convolution





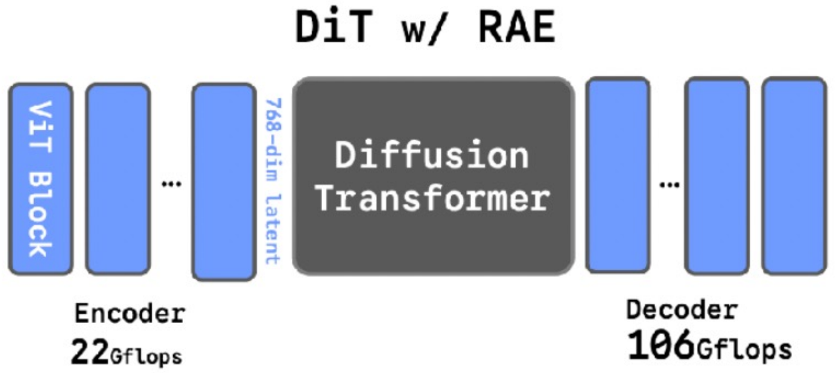
(a) 2D convolution

(b) 3D convolution

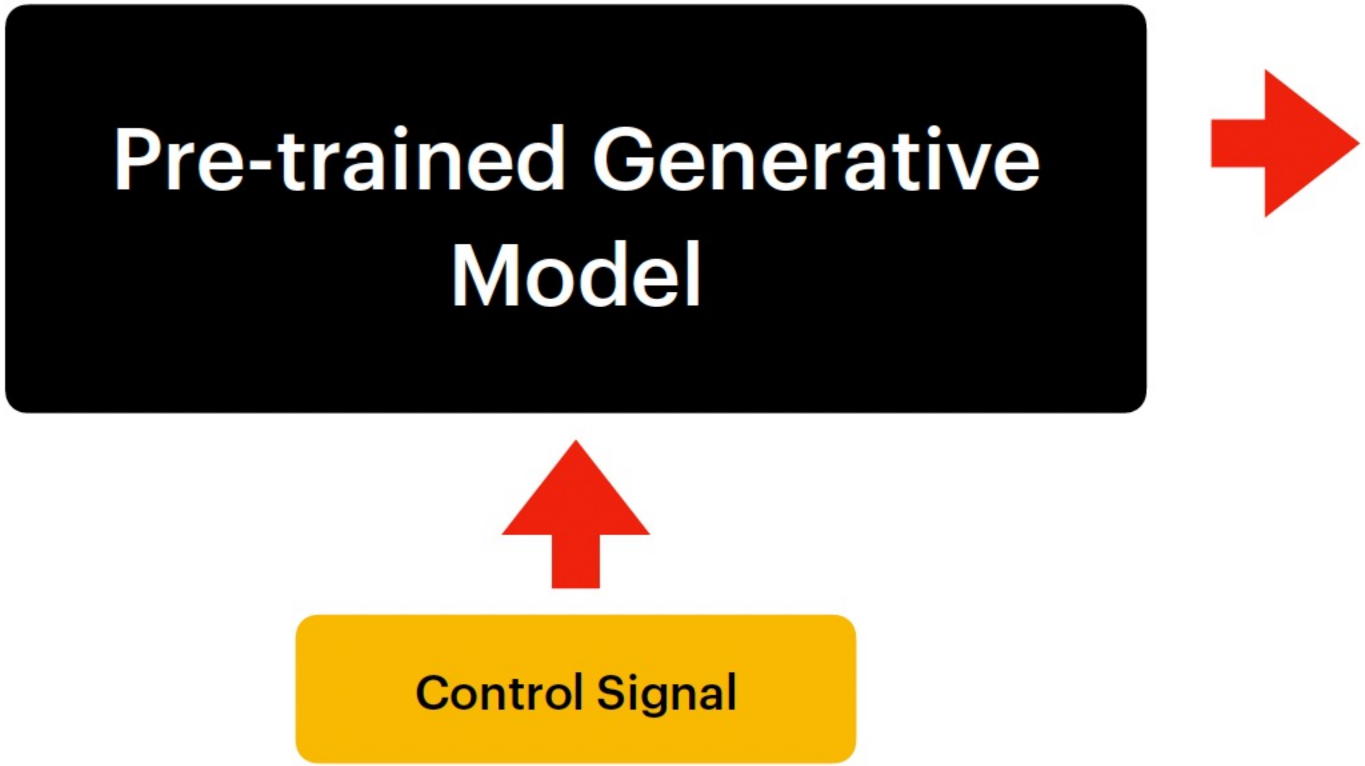
Causal Convolution —> Transformer / RNN-based VAE?

DIFFUSION TRANSFORMERS WITH REPRESENTATION AUTOENCODERS

Boyang Zheng Nanye Ma Shengbang Tong Saining Xie
New York University



Controllable Generation



Editing

Personalization

Guidance

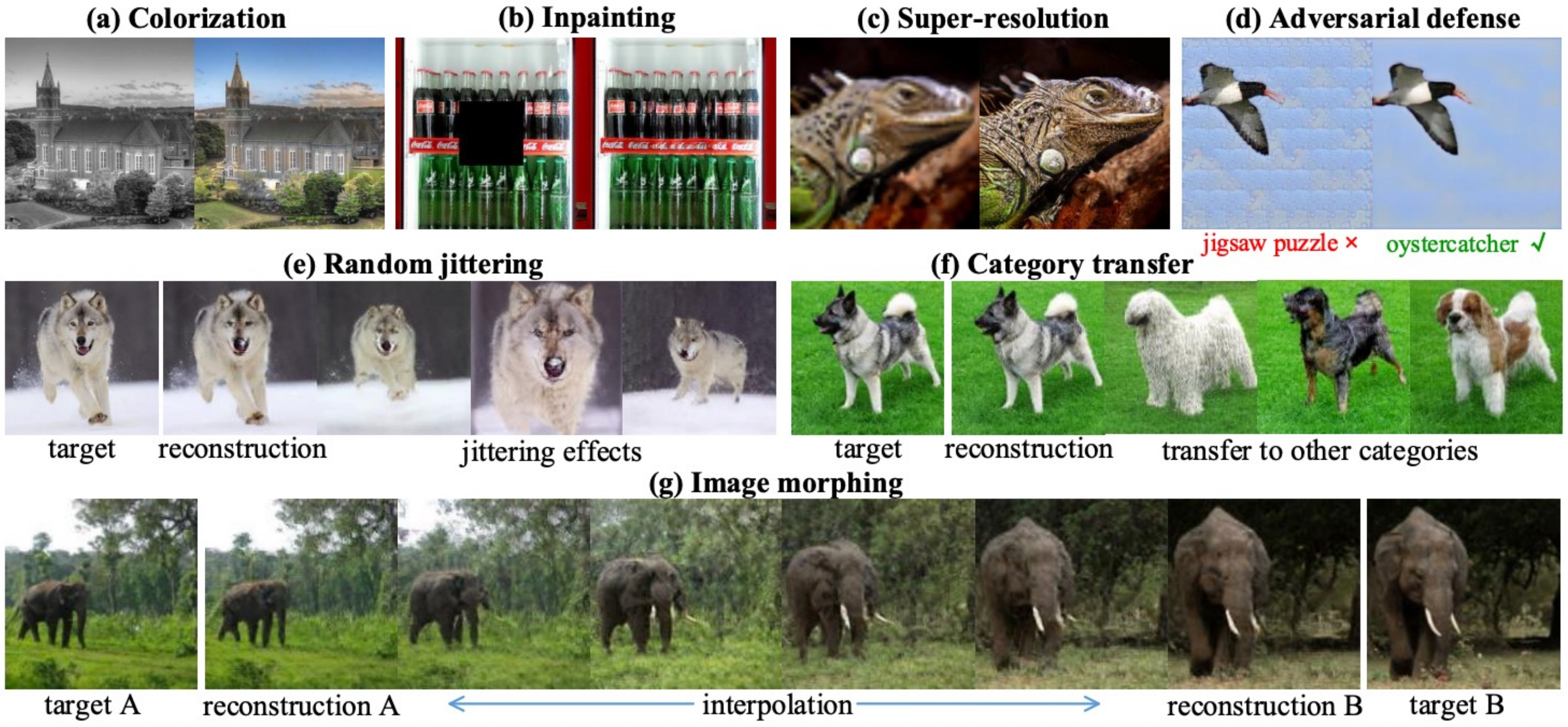
...

Exploiting Deep Generative Prior for Versatile Image Restoration and Manipulation

Xingang Pan[1], Xiaohang Zhan[1], Bo Dai[1], Dahua Lin[1], Chen Change Loy[2], and Ping Luo[3]

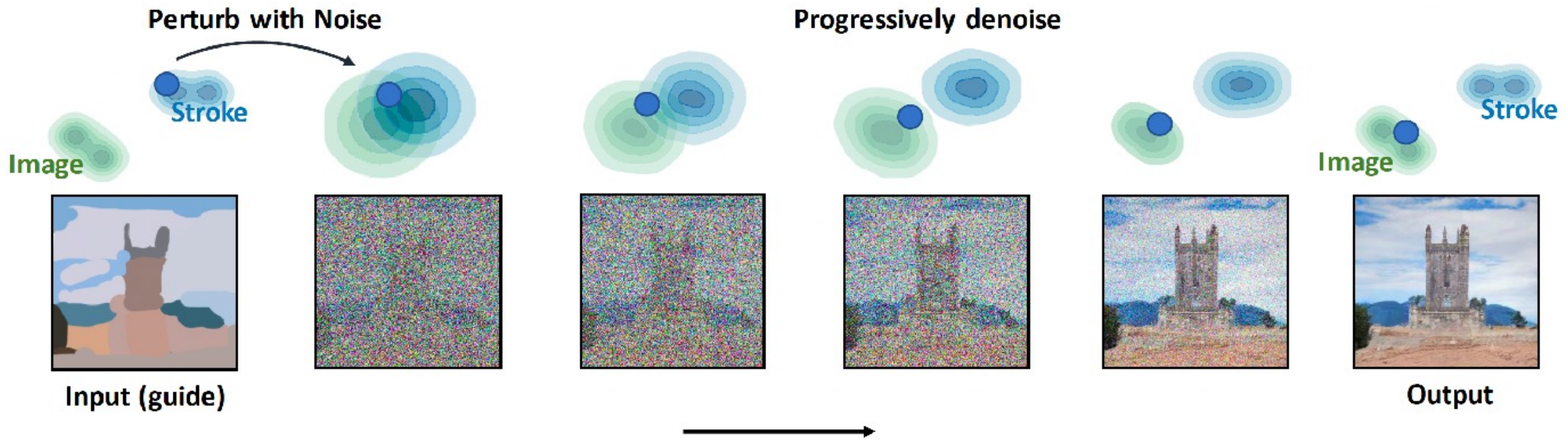
$$\mathbf{z}^* = \arg\min_{\mathbf{z}\in\mathbb{R}^d} E(\hat{\mathbf{x}}, G(\mathbf{z};\boldsymbol{\theta})), \qquad \mathbf{x}^* = G(\mathbf{z}^*;\boldsymbol{\theta}),$$

$$= \arg\min_{\mathbf{z}\in\mathbb{R}^d} \mathcal{L}(\hat{\mathbf{x}}, \phi(G(\mathbf{z};\boldsymbol{\theta}))),$$



SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations

First perturb the input with **Gaussian noise** and then progressively remove the noise using a pretrained diffusion model.

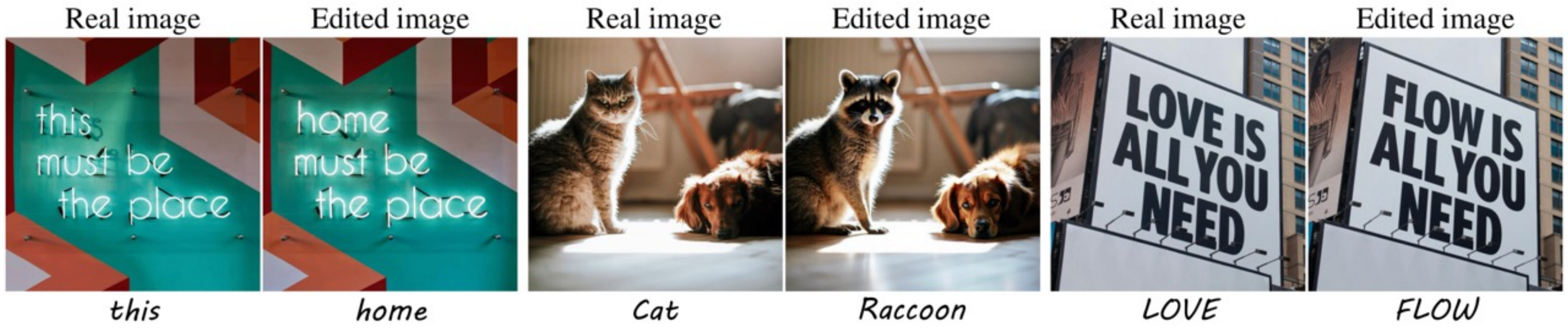


Gradually projects the input to the manifold of natural images.

FlowEdit: Inversion-Free Text-Based Editing Using Pre-Trained Flow Models

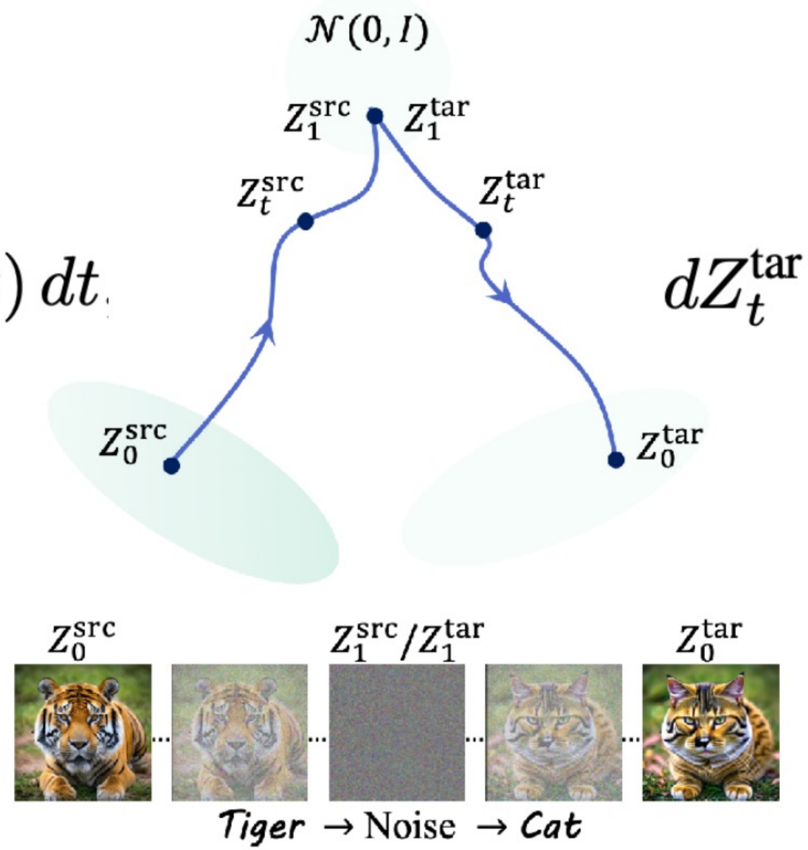
ICCV 2025 Best Student Paper

Vladimir Kulikov Matan Kleiner Inbar Huberman-Spiegelglas Tomer Michaeli
Technion – Israel Institute of Technology



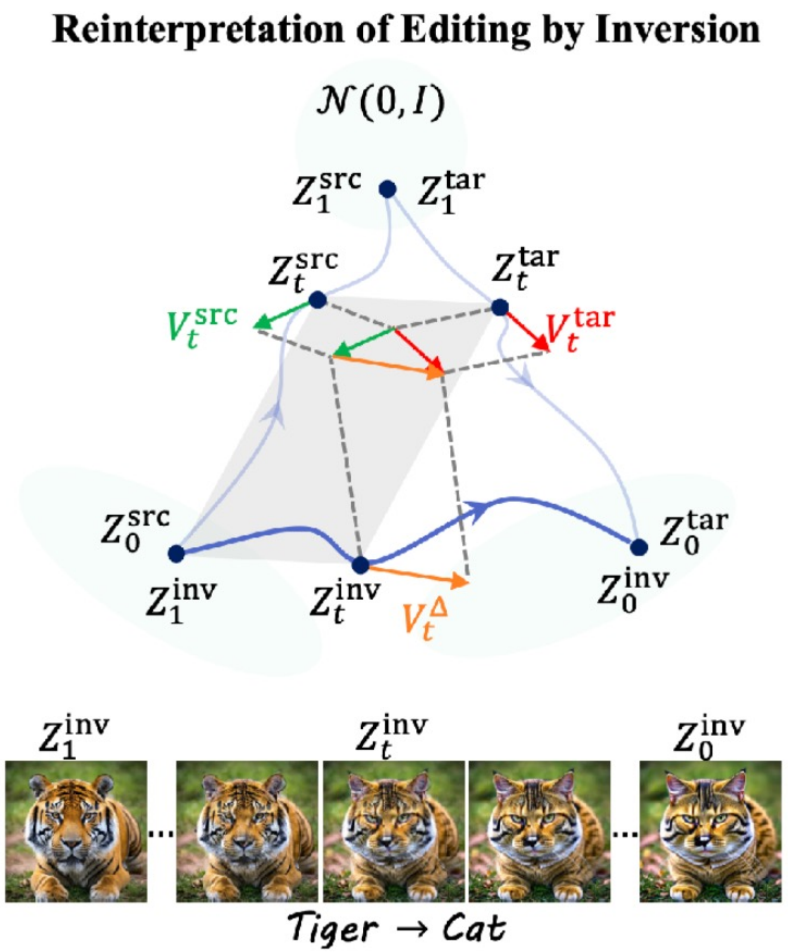
$$dZ_t^{\text{src}} = V^{\text{src}}(Z_t^{\text{src}}, t)\, dt$$

Editing by ODE Inversion



$$dZ_t^{\text{tar}} = V^{\text{tar}}(Z_t^{\text{tar}}, t)\, dt$$

Controllable Generation: Inversion-based



Define: $Z_t^{\text{inv}} = Z_0^{\text{src}} + Z_t^{\text{tar}} - Z_t^{\text{src}}$

$$\Downarrow$$

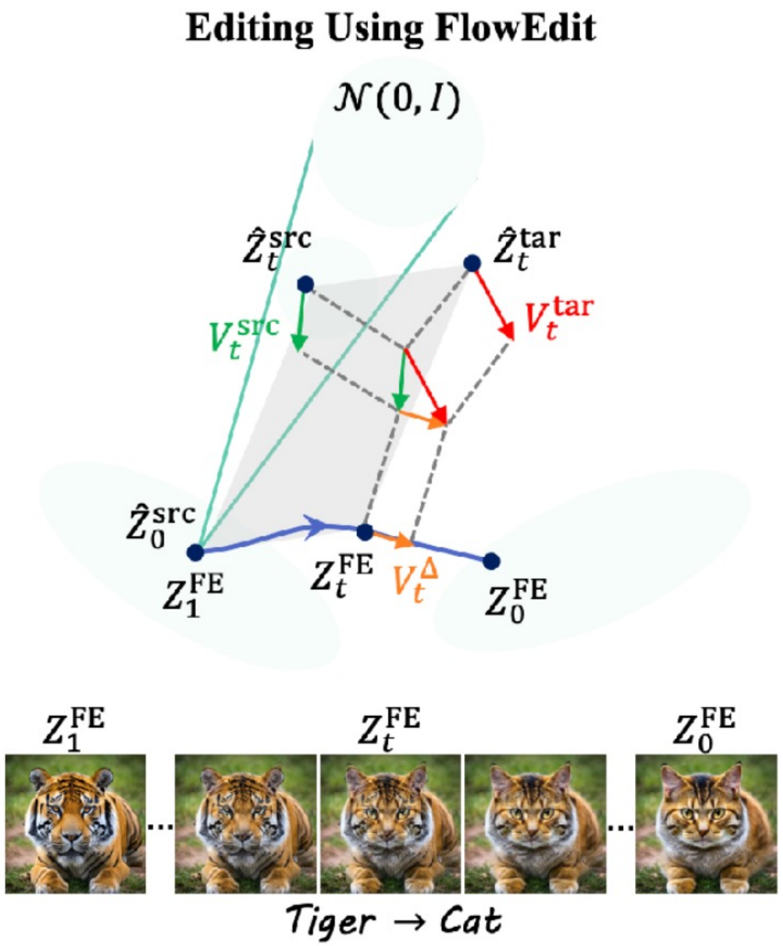
$$dZ_t^{\text{inv}} = V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}})\, dt$$

$$V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}}) = V^{\text{tar}}(Z_t^{\text{tar}}, t) - V^{\text{src}}(Z_t^{\text{src}}, t)$$

$$\Downarrow$$

$$dZ_t^{\text{inv}} = V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{inv}} + Z_t^{\text{src}} - Z_0^{\text{src}})\, dt$$

(Noise-free path)



Find a "shorter path" to the nearest mode

$$dZ_t^{\text{inv}} = V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{inv}} + Z_t^{\text{src}} - Z_0^{\text{src}})\, dt$$

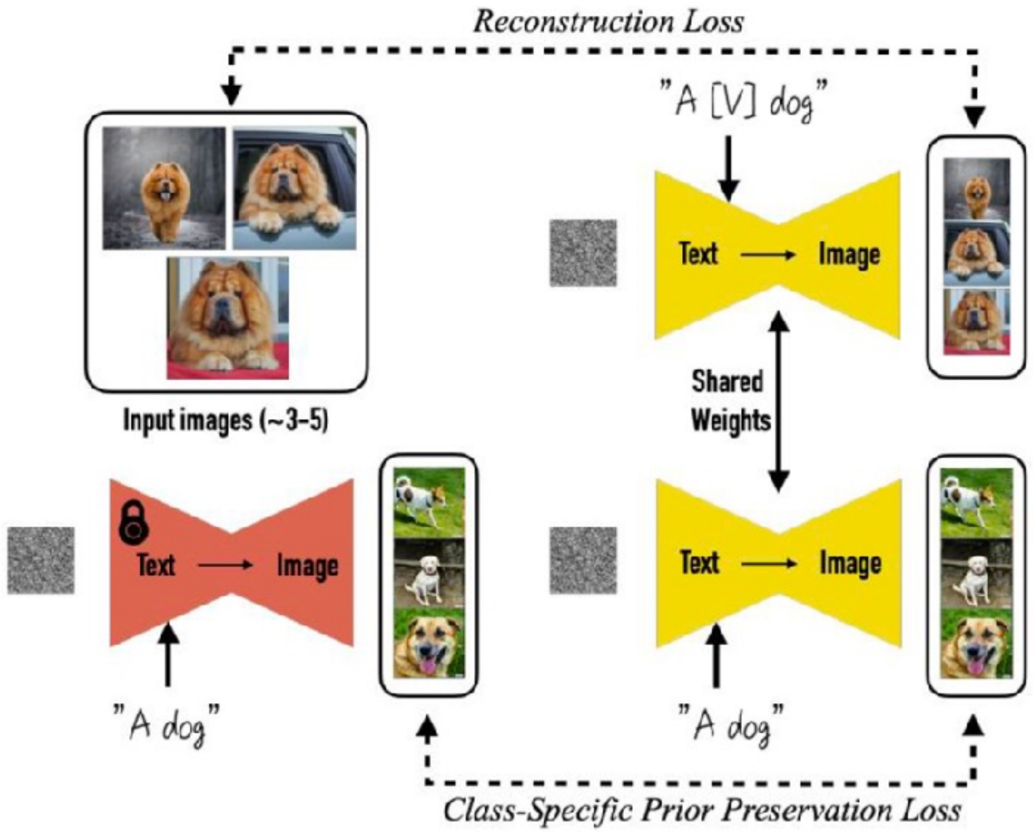
(Reusing the ODE, but not using inversion)

$$\hat{Z}_t^{\text{src}} = (1-t)Z_0^{\text{src}} + tN_t,$$

(Sample N pairs, and take average velocity field)

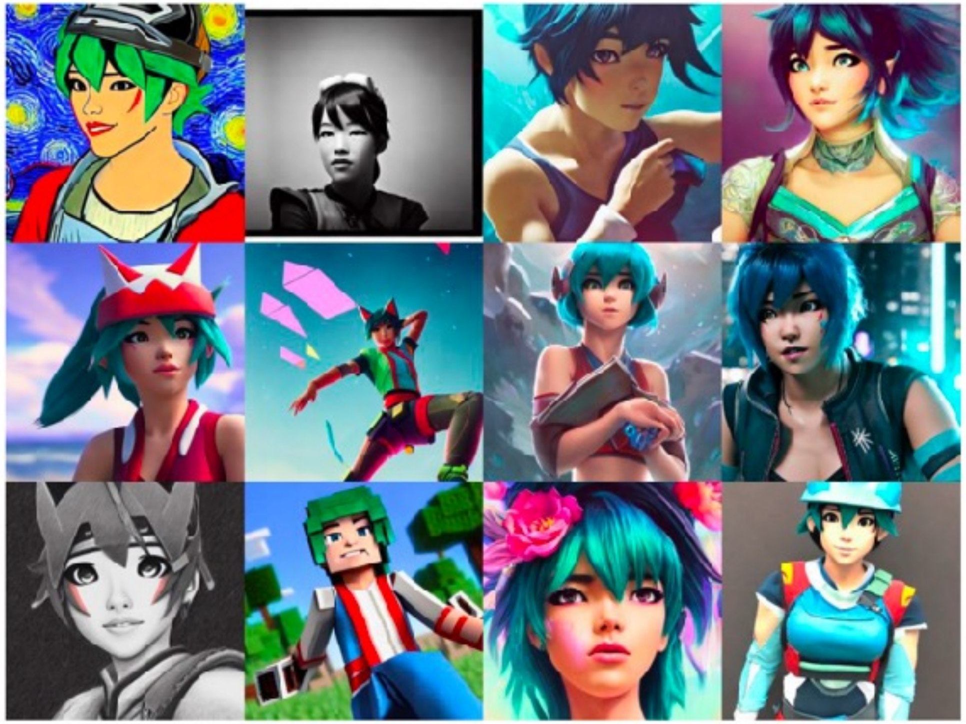
$$dZ_t^{\text{FE}} = \mathbb{E}\left[V_t^{\Delta}(\hat{Z}_t^{\text{src}}, Z_t^{\text{FE}} + \hat{Z}_t^{\text{src}} - Z_0^{\text{src}}) \,\middle|\, Z_0^{\text{src}}\right] dt$$

DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation



Low-rank Adaptation (LoRA)

- Lora: Low-rank adaptation of large language models

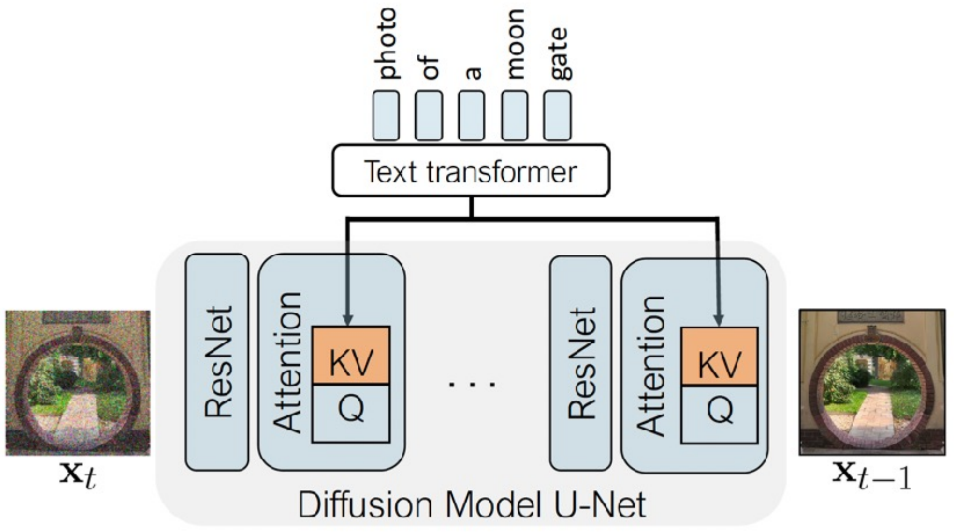


Original weights

$$W = \underbrace{W_0}_{\text{Original weights}} + \underbrace{BA}_{\text{Low-rank difference}}$$

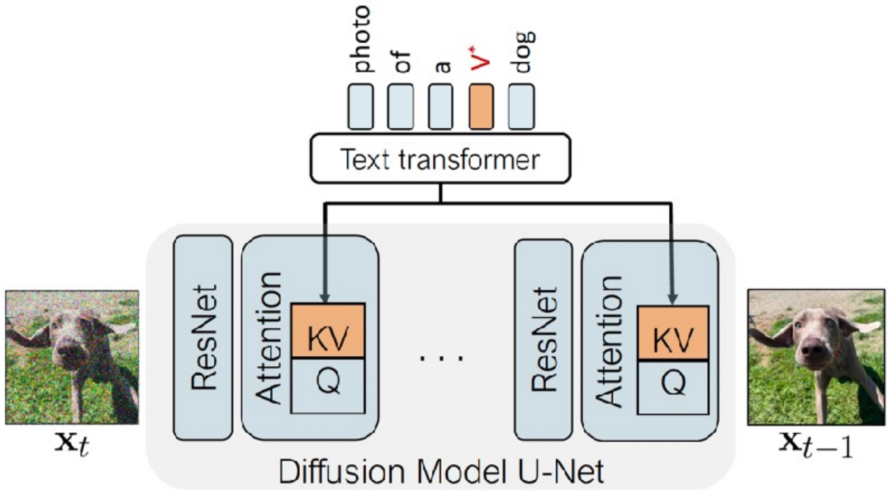
Low-rank difference

Only fine-tune cross-attention layers

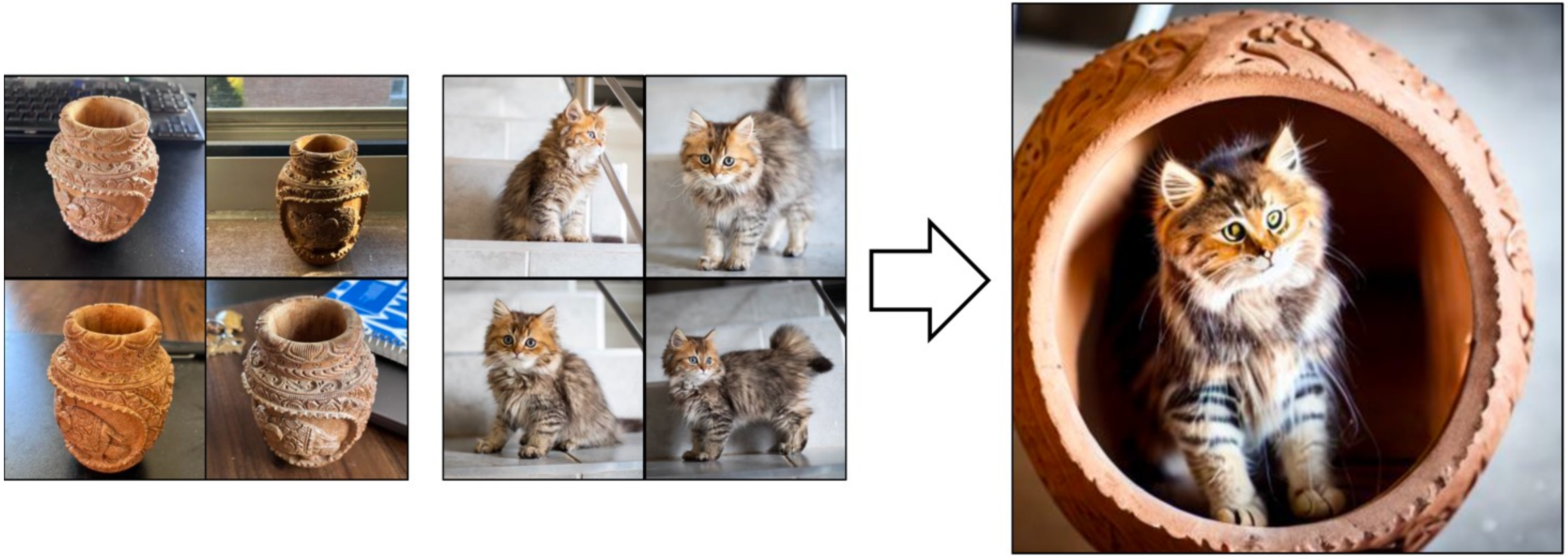


Personalized concepts

Also fine-tune the modifier token V^* that describes the personalized concept

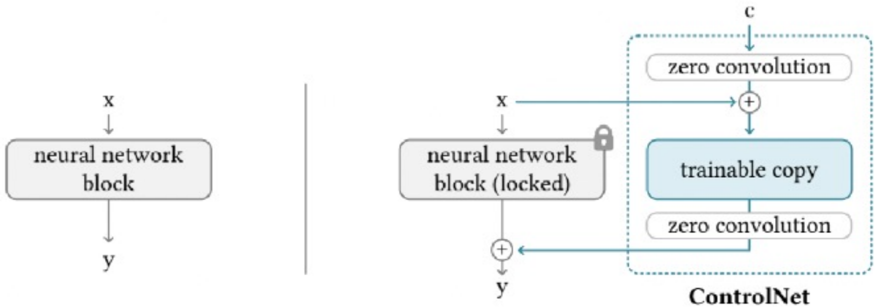


Two concept results



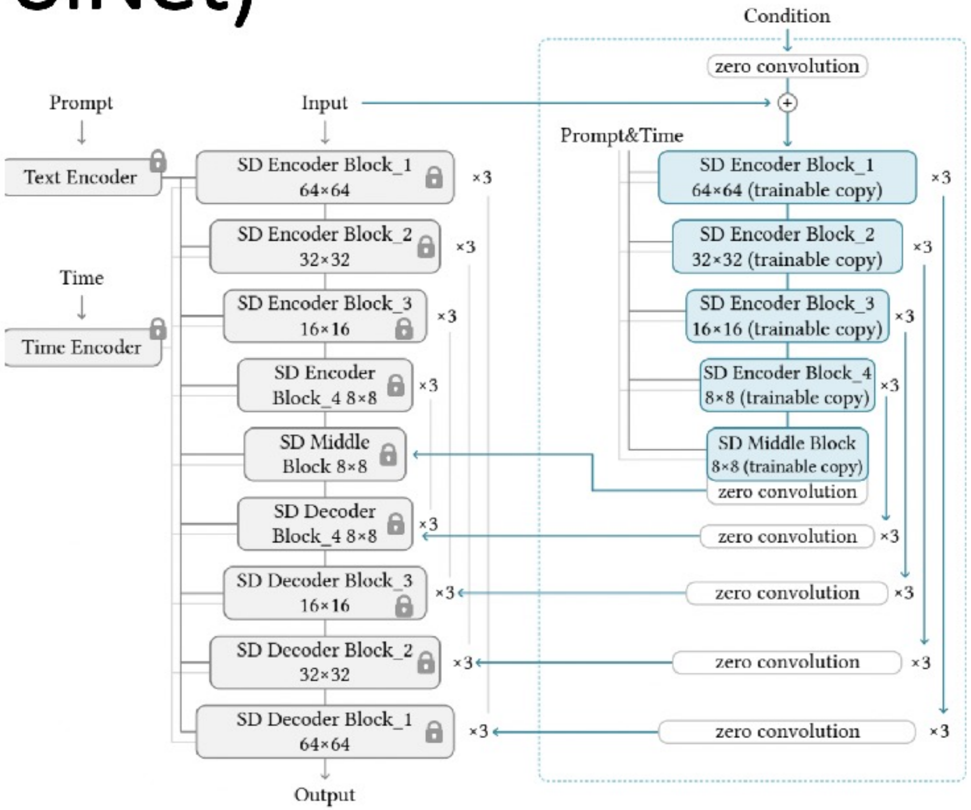
The V_1^* cat is sitting inside a V_2^* wooden pot and looking up

Adding Conditional Control to Text-to-Image Diffusion Models (ControlNet)



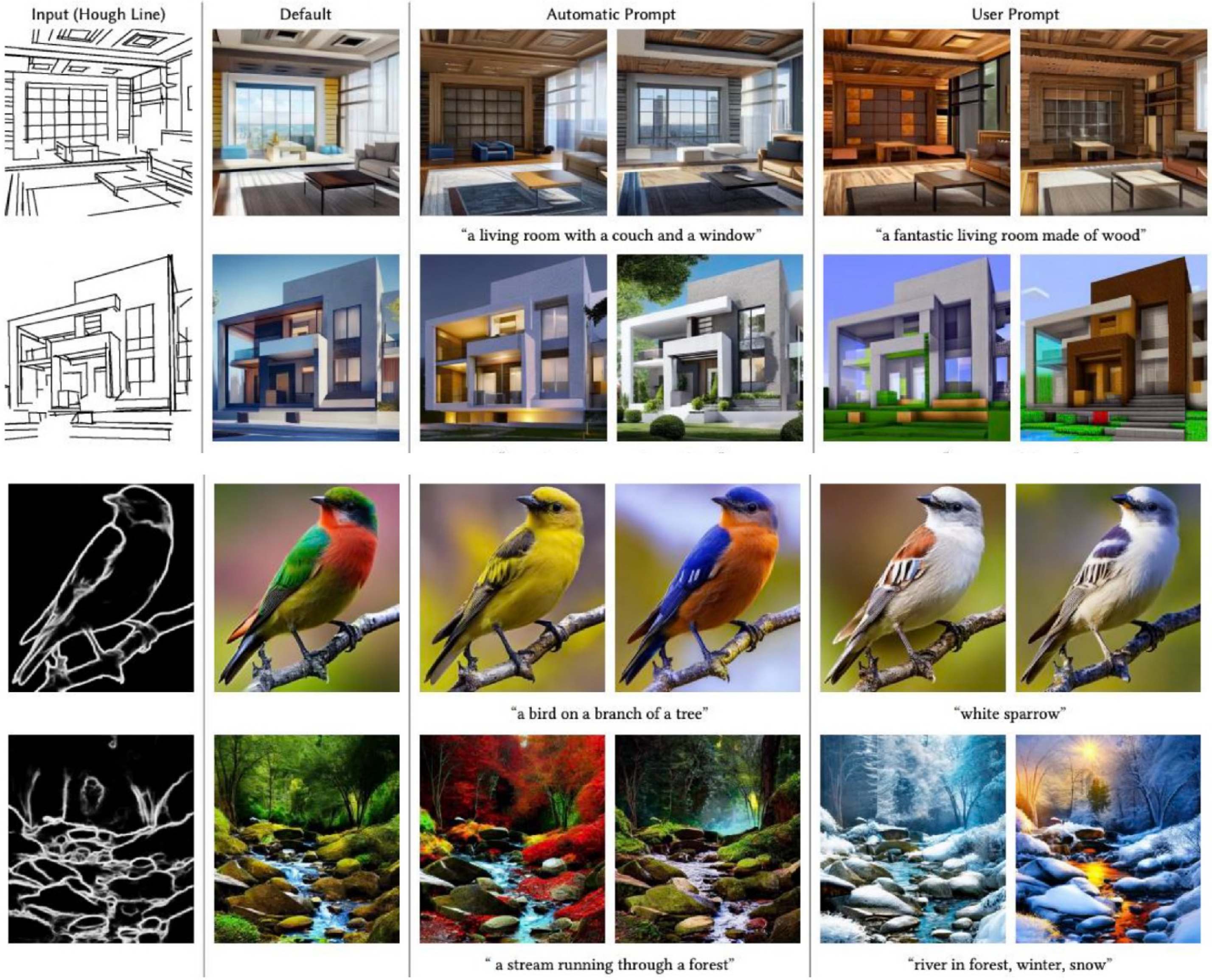
(a) Before (b) After

ControlNet



(a) Stable Diffusion (b) ControlNet

Controllable Generation: Module-based

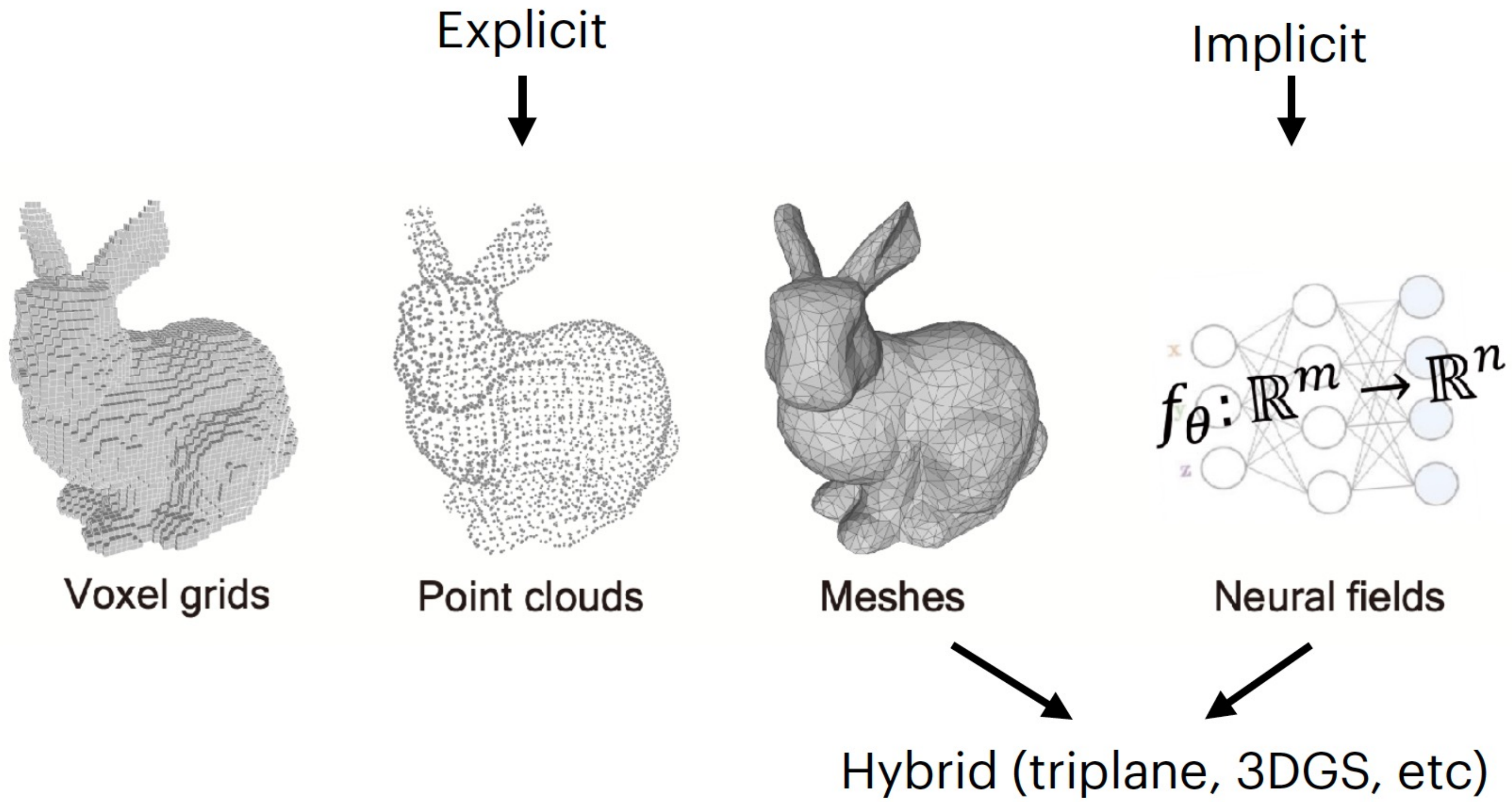


How to represent 3D?

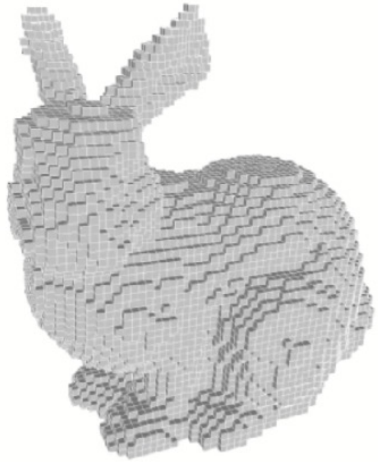
Images/Videos —> Pixel grids

Voxel grids

How to represent 3D?



How to represent 3D?


Voxel grids

Pros:

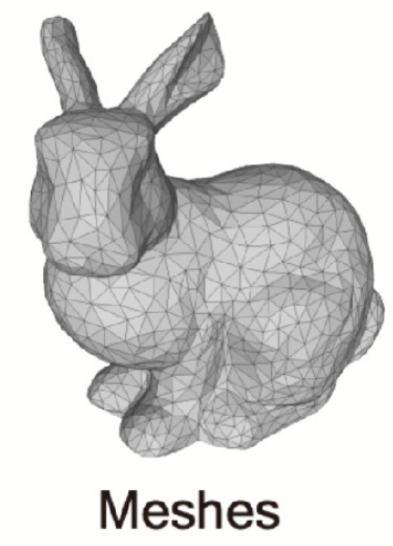
- Similar to Pixels, very suitable for NNs

Cons:

- Limited resolution (N^3)
- Low memory efficiency; Slow

Our space is vastly empty!

How to represent 3D?



Pros:

- Only model the surface, very efficient
- Commonly used in the CG industry

Cons:

- Irregular (vertex, edges), not very suitable for NNs
- Not flexible

How to represent 3D?

Point clouds

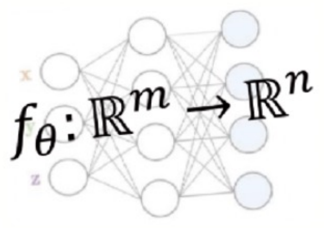
Pros:

- Flexible; Seen as samples from the density
- Easier to handle (e.g., Transformers)

Cons:

- Less efficient than Meshes
- Not continuous; Stochastic

How to represent 3D?



Neural fields

Pros:

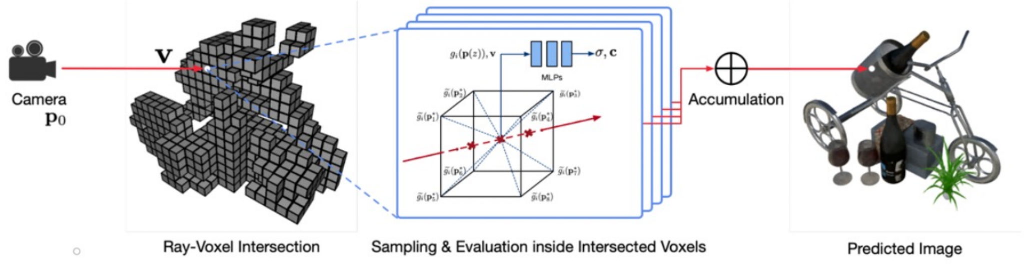
- Flexible; Infinite resolution
- Compact; memory efficient

Cons:

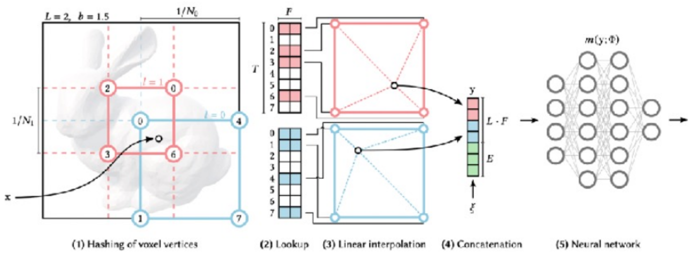
- Querying NNs for every point is extremely slow
- Fields by itself is hard to handle by NNs (e.g., HyperNetworks?)

How to represent 3D?

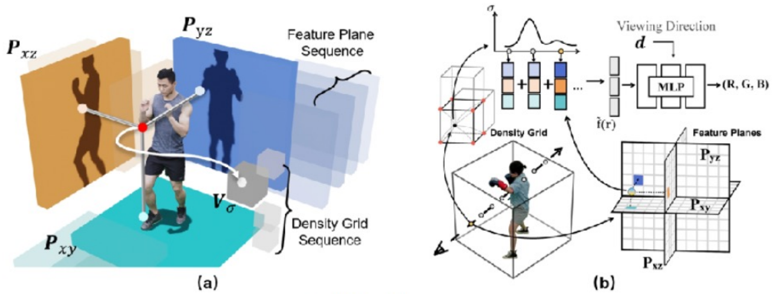
Explicit + Implicit Hybrid => Better efficiency and more balanced



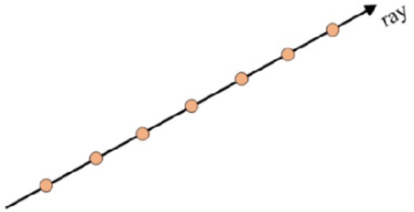
(Sparse) Voxels



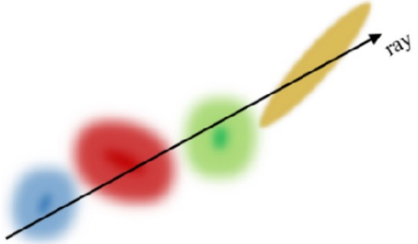
Hashtables



Triplanes



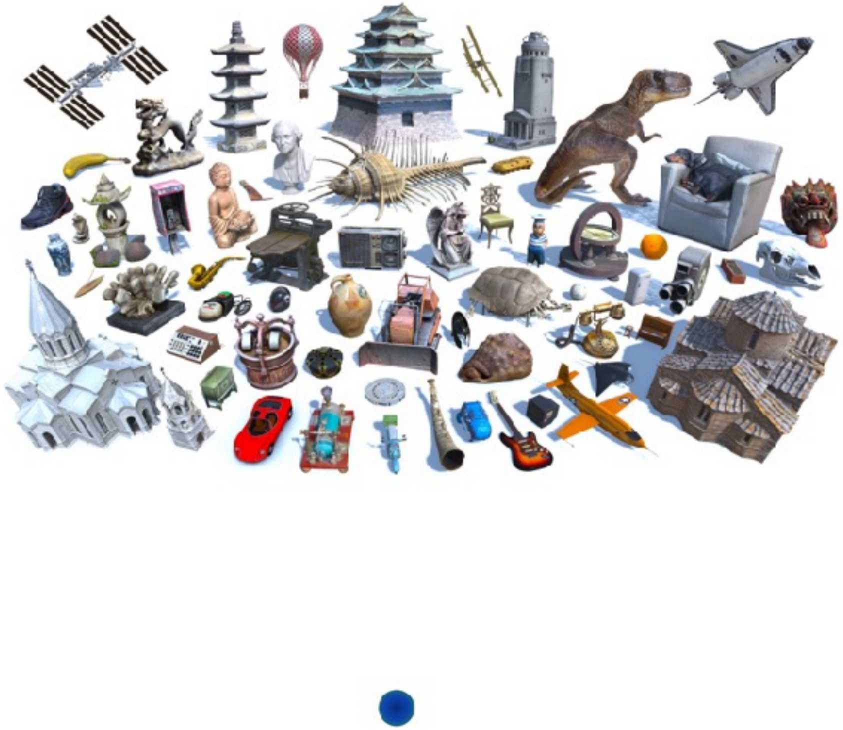
NeRF



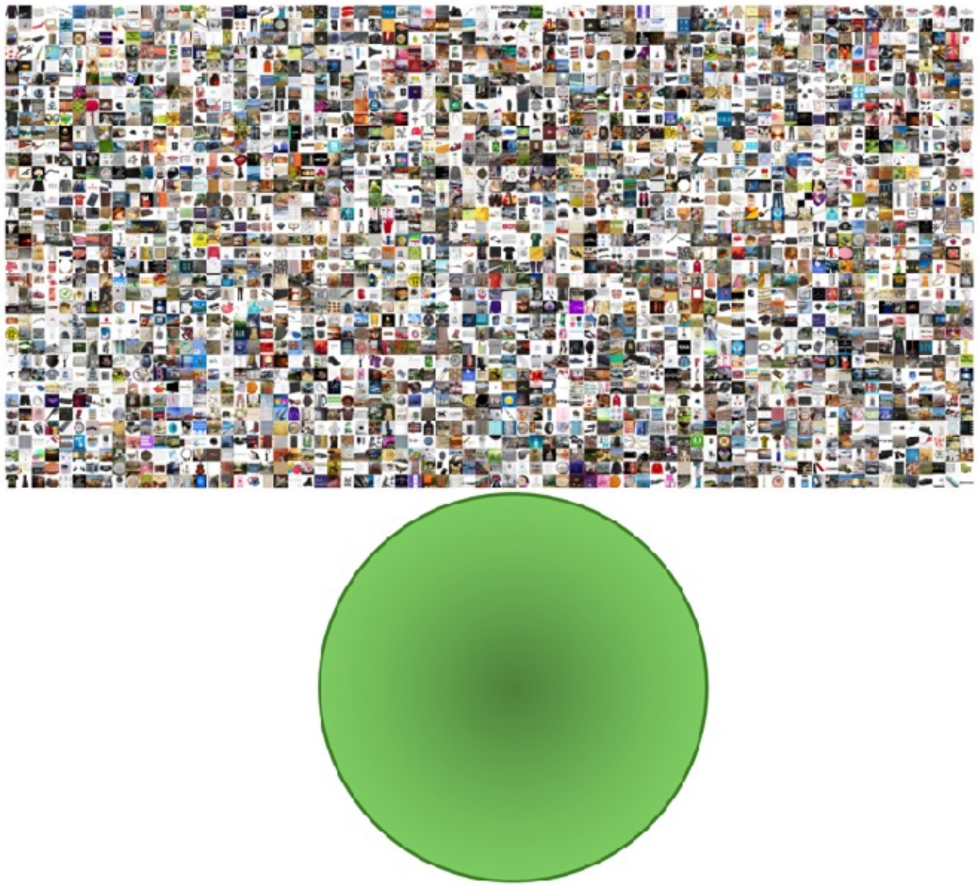
Gaussian Splatting

How to represent 3D? Why so many types of representations?

- 3D is not a unified concept, containing heterogenous types of content



Objaverse [1]
(800K)

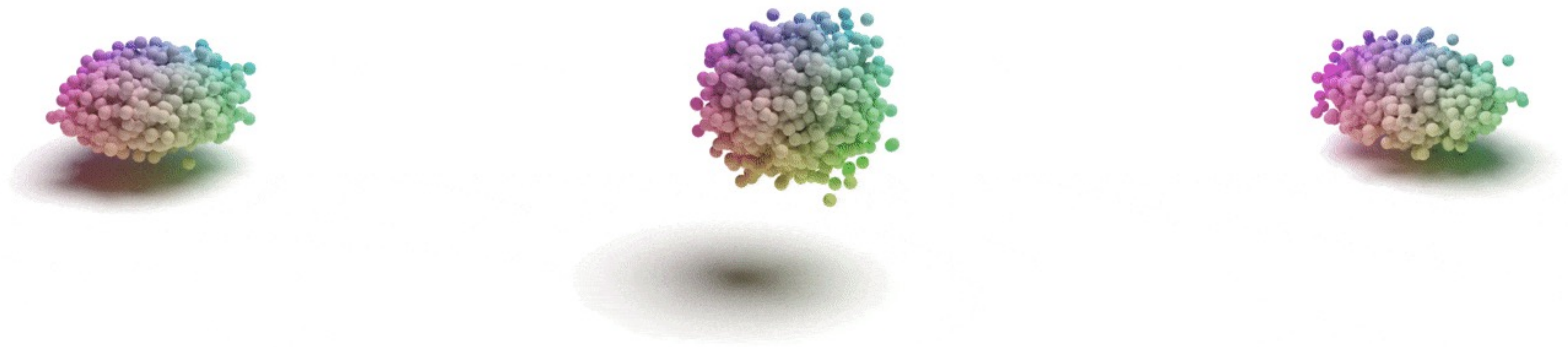


Coyo [2]
(700M)

HKU Musketeers Foundation Institute of Data Science 香港大學同心基金數據科學研究院

PointFlow: 3D Point Cloud Generation with Continuous Normalizing Flows

Guandao Yang[1,2*], Xun Huang[1,2*], Zekun Hao[1,2], Ming-Yu Liu[3], Serge Belongie[1,2], Bharath Hariharan[1]
[1]Cornell University [2]Cornell Tech [3]NVIDIA

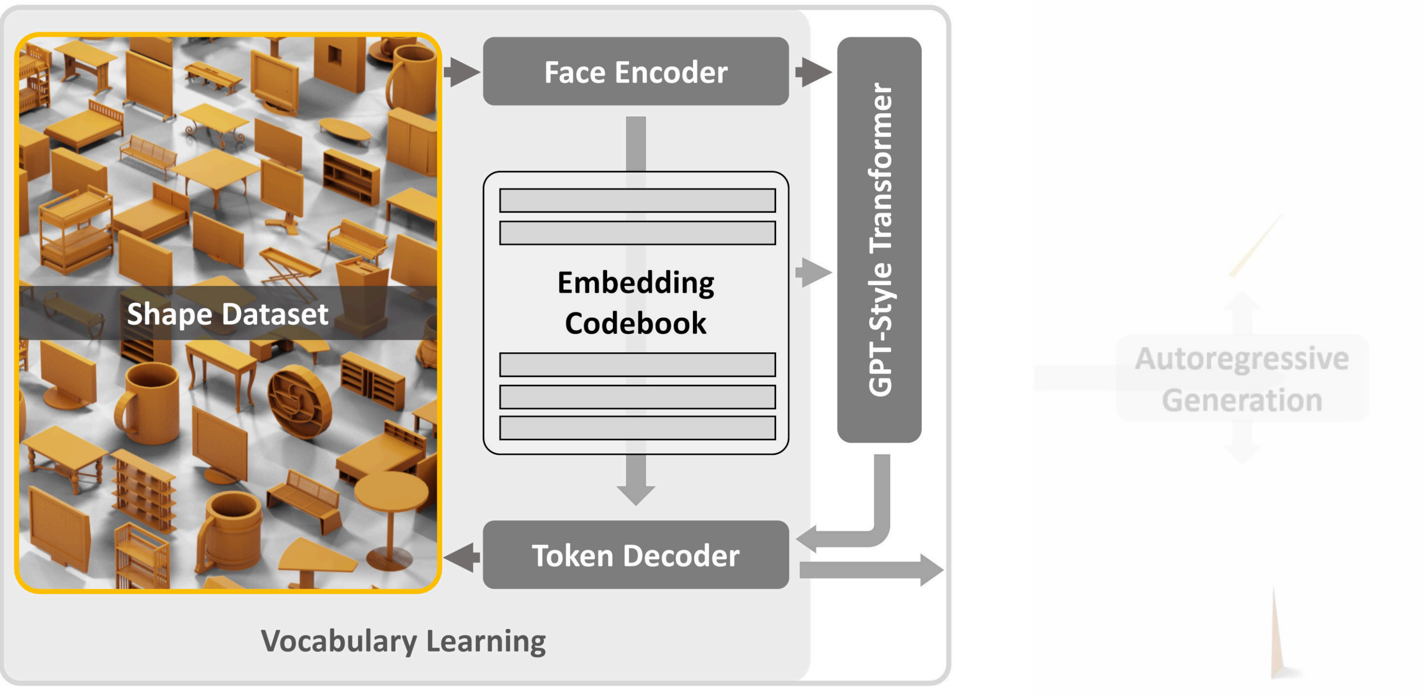


MeshGPT: Generating Triangle Meshes with Decoder-Only Transformers

CVPR 2024 Highlight

Yawar Siddiqui[1], Antonio Alliegro[2], Alexey Artemov[1], Tatiana Tommasi[2], Daniele Sirigatti[3], Vladislav Rosov[3], Angela Dai[1], Matthias Nießner[1]

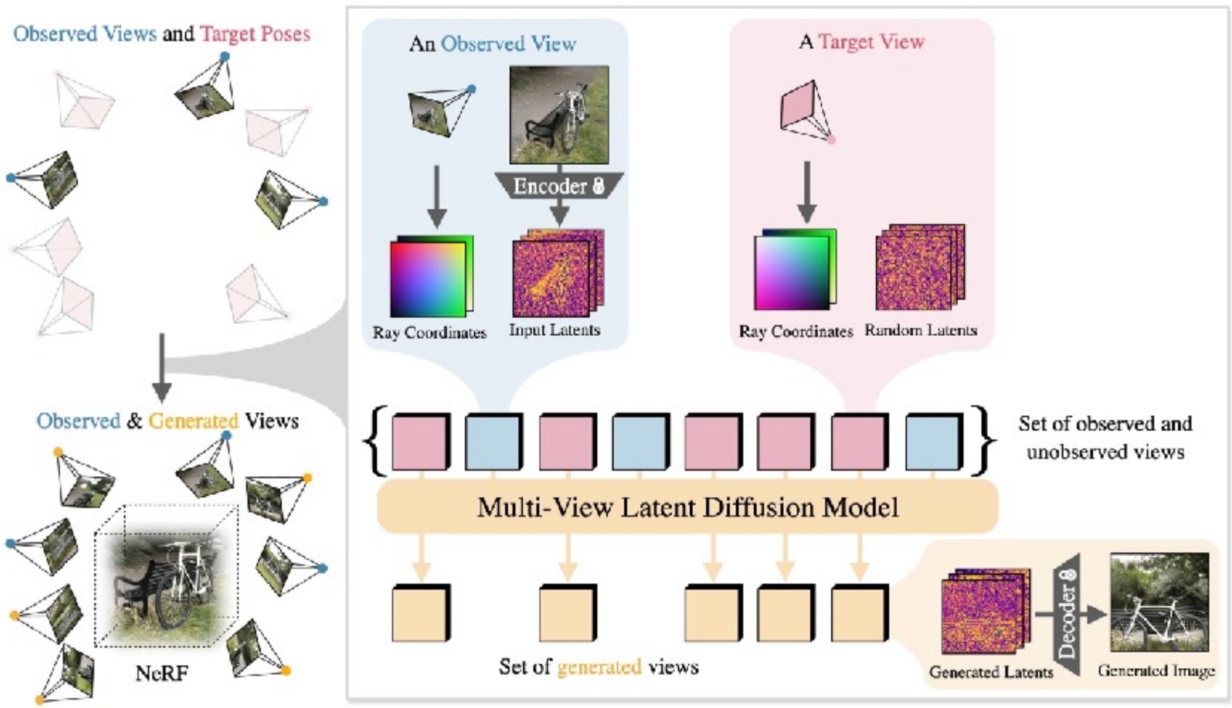
[1]Technical University of Munich, [2]Politecnico di Torino, [3]AUDI AG



CAT3D: Create Anything in 3D with Multi-View Diffusion Models

Ruiqi Gao[1]* Aleksander Holynski[1]*
Philipp Henzler[2] Arthur Brussee[1] Ricardo Martin-Brualla[2]
Pratul P. Srinivasan[1] Jonathan T. Barron[1] Ben Poole[1]*
[1]Google DeepMind [2]Google Research *equal contribution

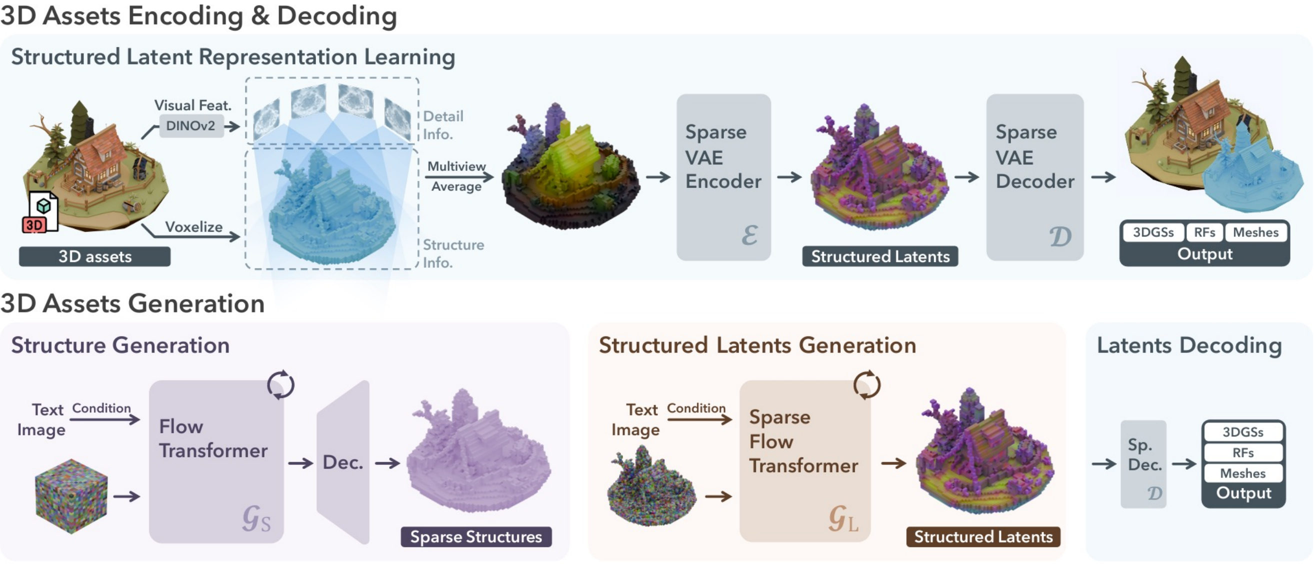
NeurIPS 2024 (Oral)

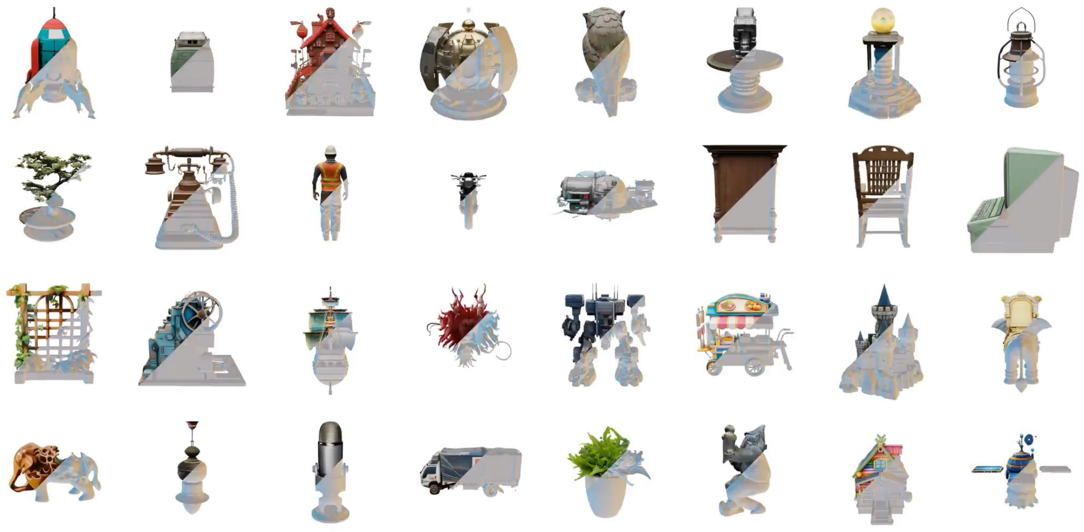


TRELLIS*

Structured 3D Latents for Scalable and Versatile 3D Generation

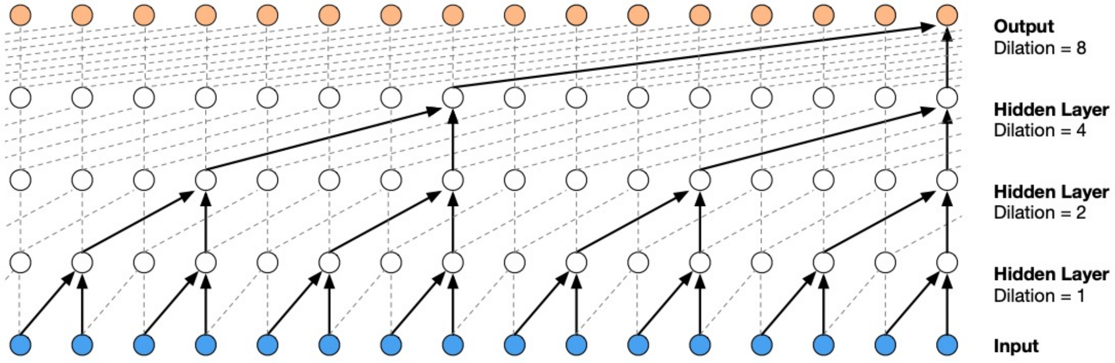
Jianfeng Xiang[1,3] Zelong Lv[2,3] Sicheng Xu[3] Yu Deng[3] Ruicheng Wang[2,3]
Bowen Zhang[2,3] Dong Chen[3] Xin Tong[3] Jiaolong Yang[3]
[1]Tsinghua University [2]USTC [3]Microsoft Research

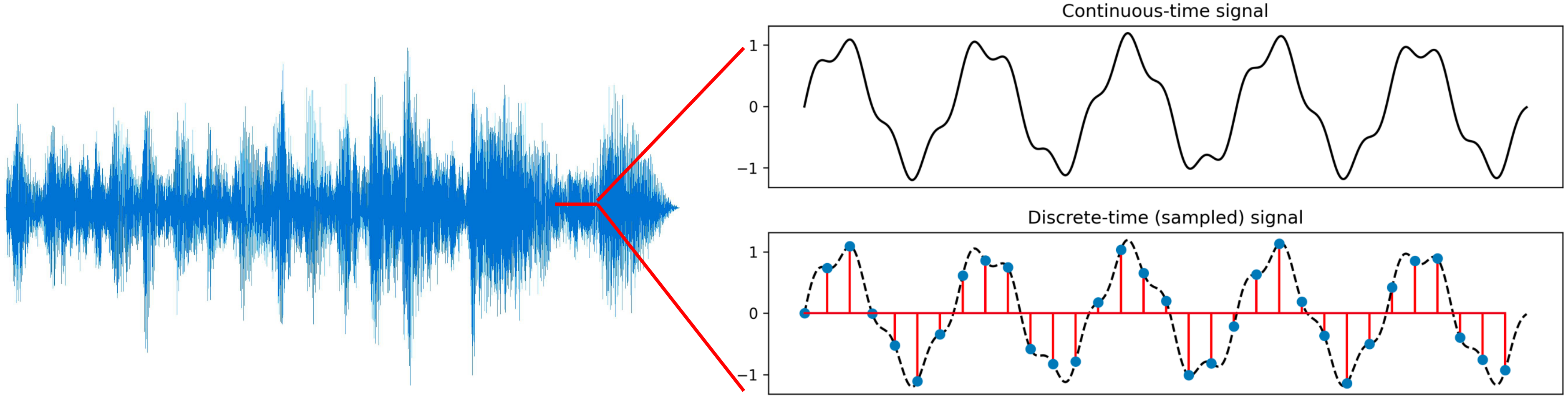




WaveNet: A Generative Model for Raw Audio

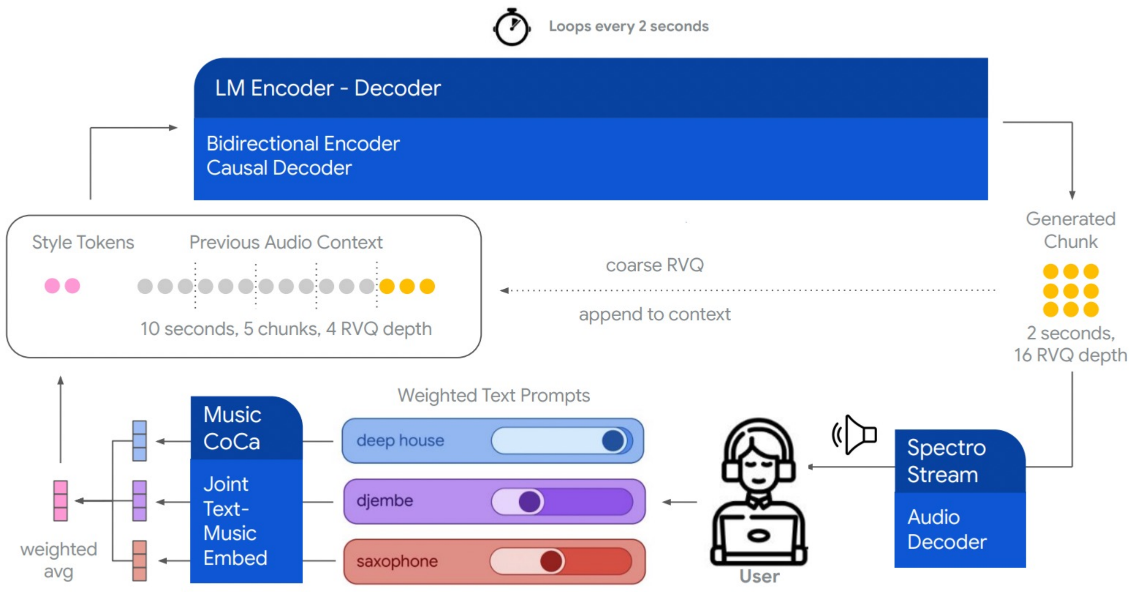
Äaron van den Oord, Sander Dieleman, Heiga Zen[†]
Karen Simonyan, Oriol Vinyals, Alex Graves
Nal Kalchbrenner, Andrew Senior, Koray Kavukcuoglu

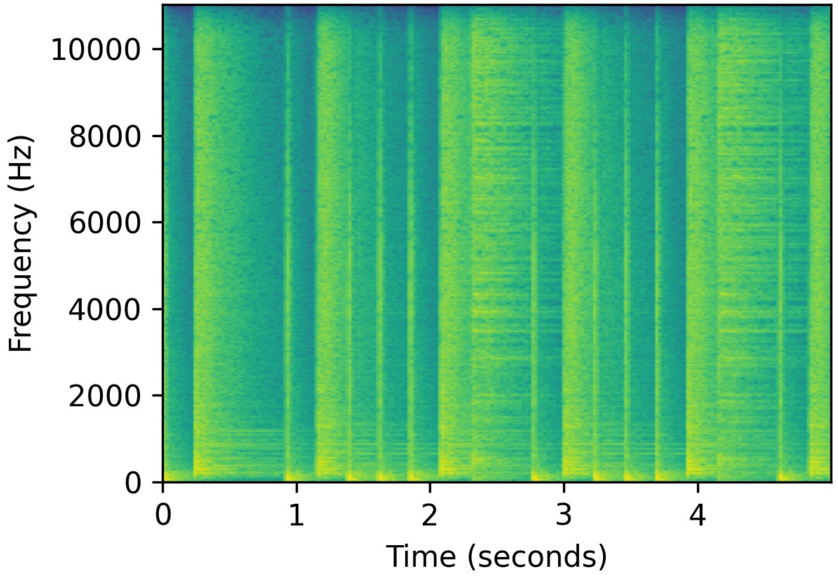


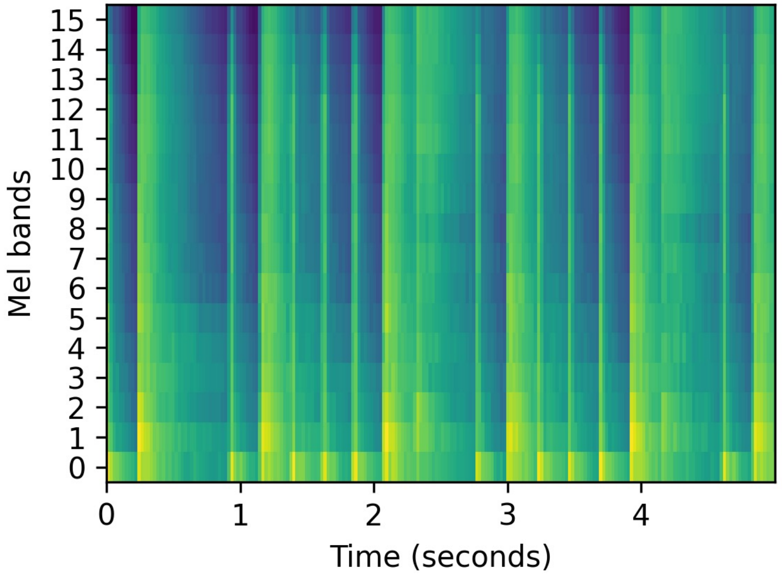


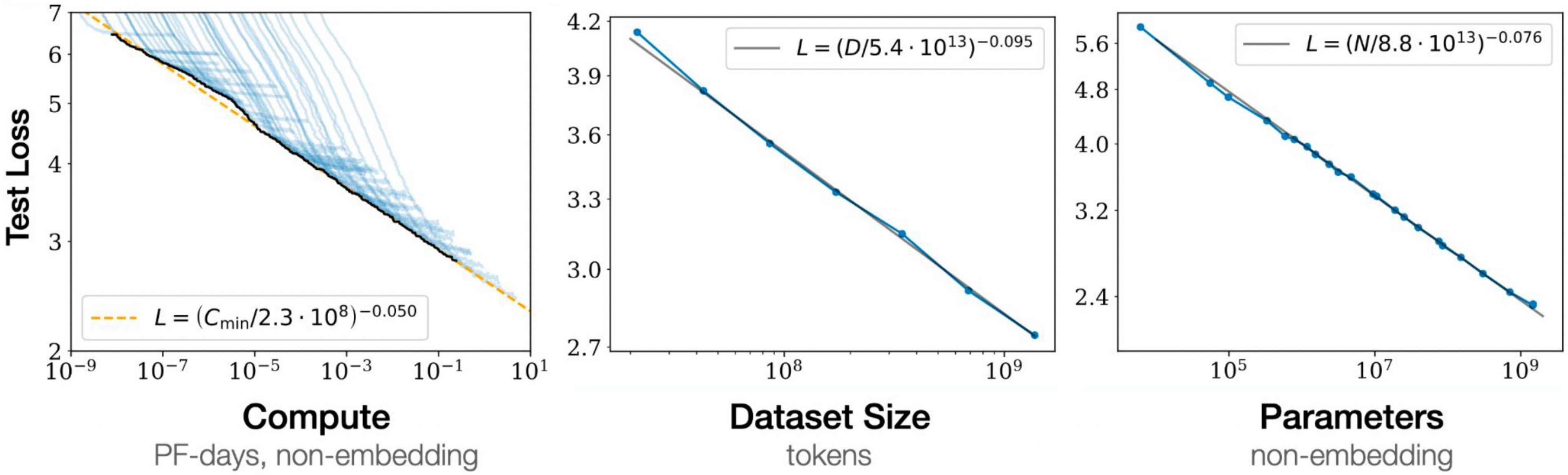
Live Music Models

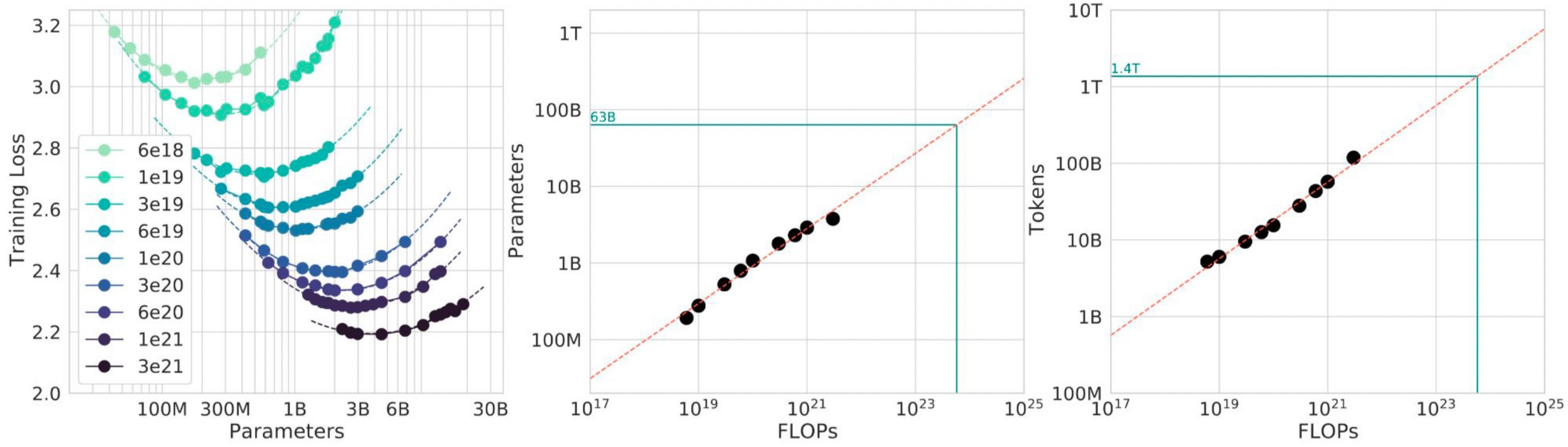
Lyria Team, Google DeepMind[1]







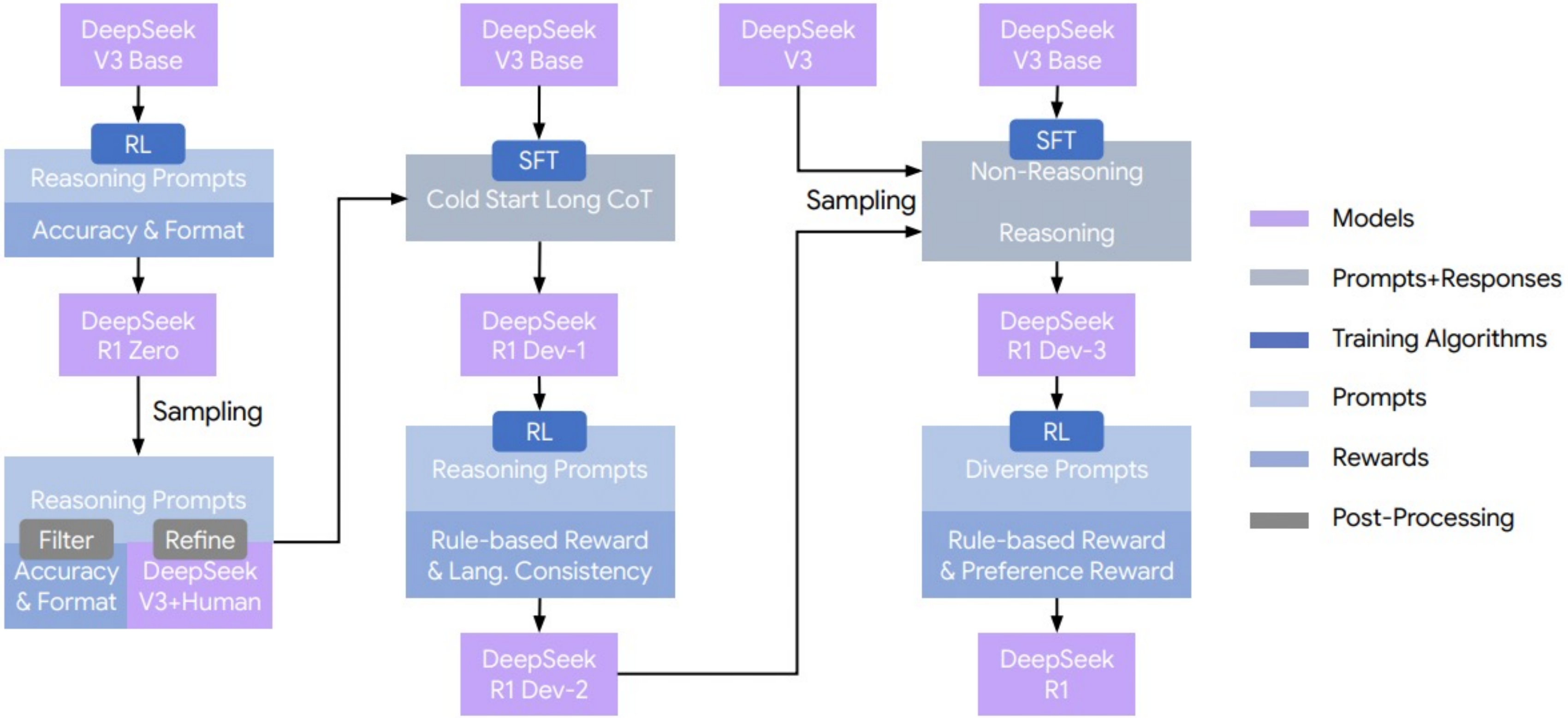

Test Loss
$L = (C_{min}/2.3 \cdot 10^8)^{-0.050}$
Compute
PF-days, non-embedding
$L = (D/5.4 \cdot 10^{13})^{-0.095}$
Dataset Size
tokens
$L = (N/8.8 \cdot 10^{13})^{-0.076}$
Parameters
non-embedding


Training Loss
6e18
1e19
3e19
6e19
1e20
3e20
6e20
1e21
3e21
Parameters
63B
FLOPs
1.4T
Tokens
FLOPs

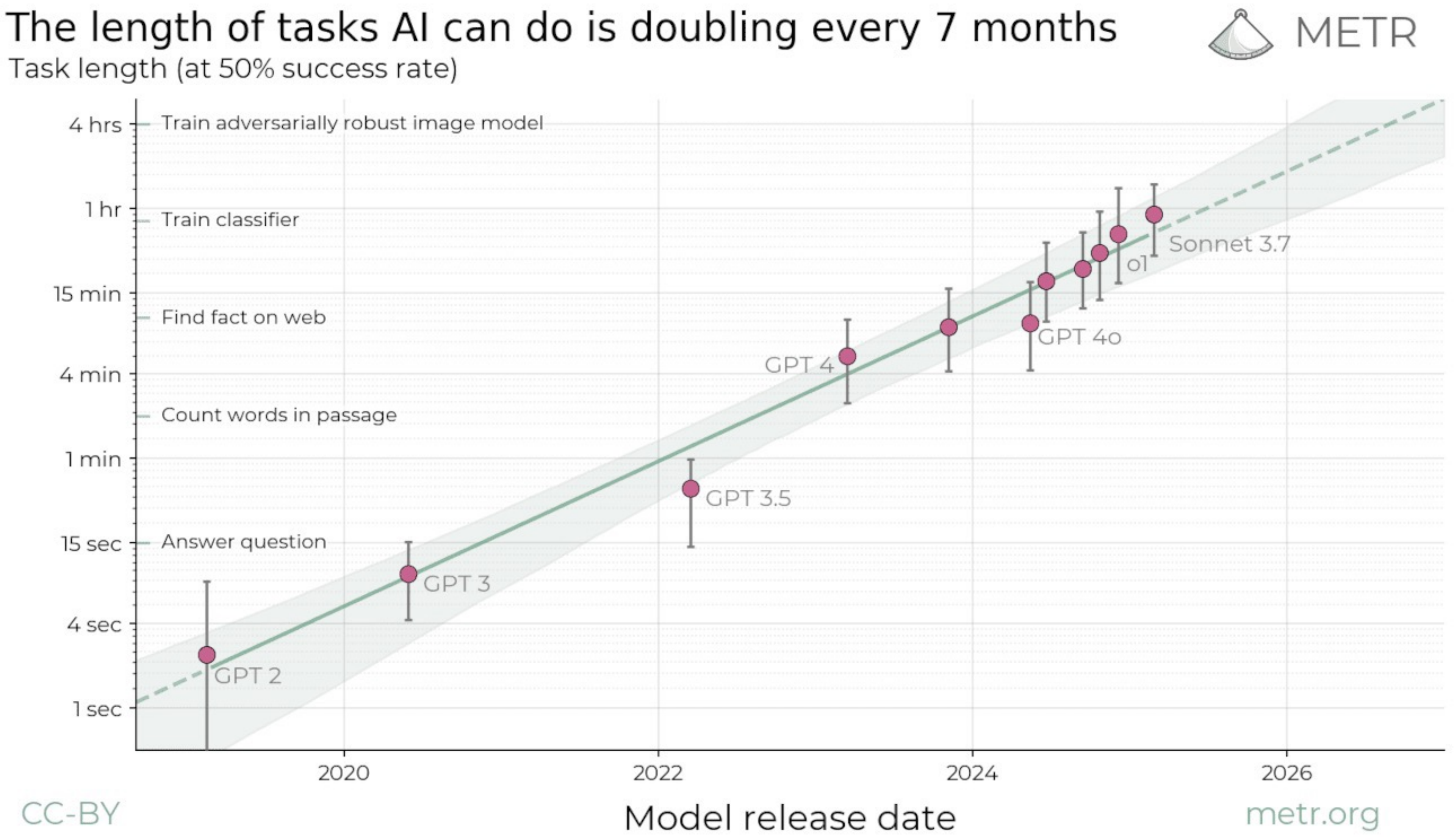
DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com



HKU Musketeers Foundation Institute of Data Science 香港大學同心基金數據科學研究院



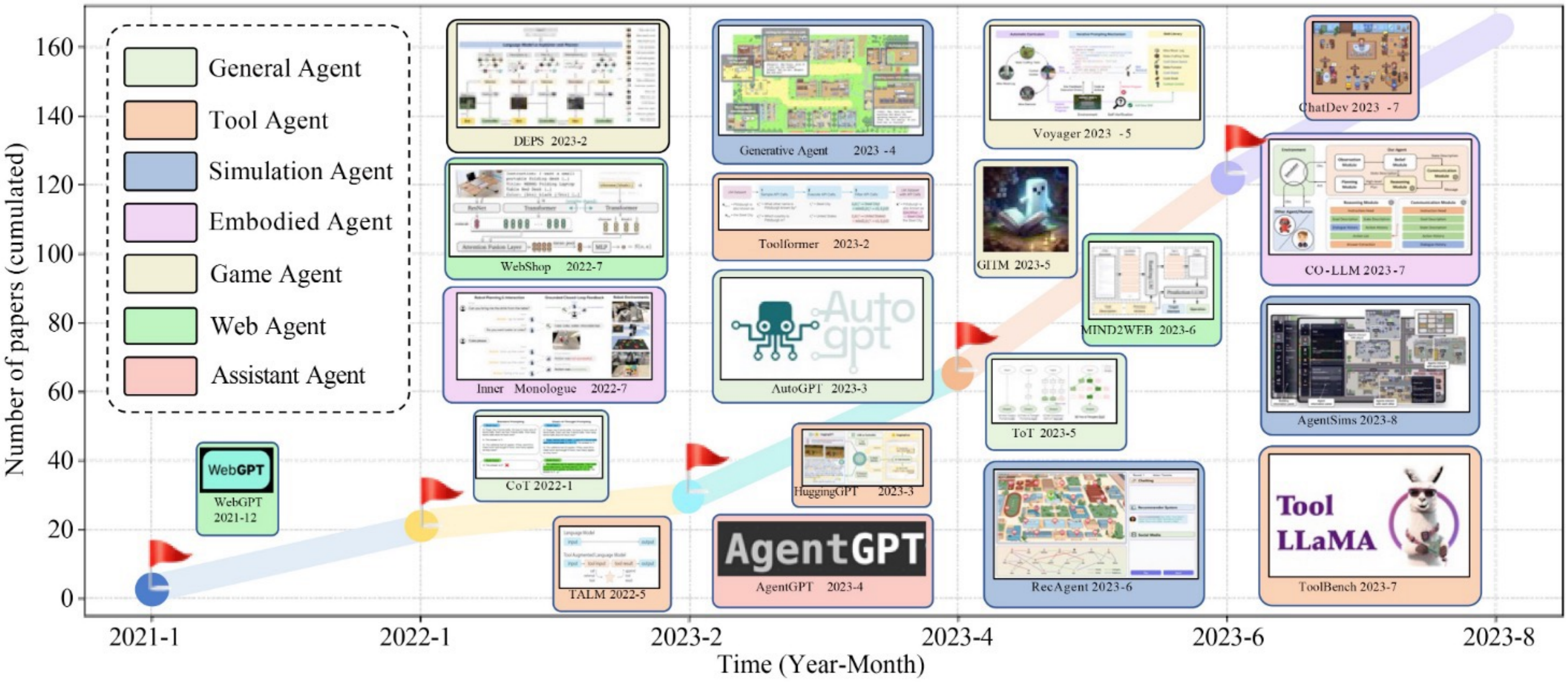
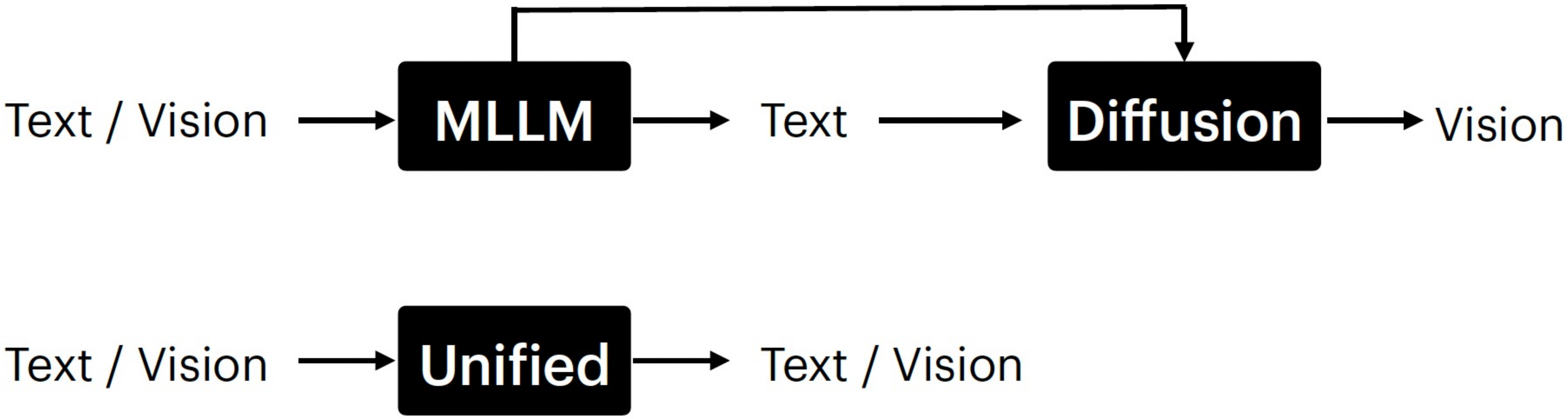


Fig. 1 Illustration of the growth trend in the field of LLM-based autonomous agents. We present the cumulative number of papers published from January 2021 to August 2023. We assign different colors to represent various agent categories. For example, a game agent aims to simulate a game-player, while a tool agent mainly focuses on tool using. For each time period, we provide a curated list of studies with diverse agent categories

Create a comic with the title: Getting started with Nano Banana Pro 🍌. Panel 1: A mobile interface on Gemini, rendered in a poetic watercolor-ink style with fine ink outlines and soft, bleeding washes. The interface is friendly and a hand painted with expressive brushwork taps a prominent button labeled "🍌 Create image". Above the button it should say "Choose your Model" then below there should be a checkbox that says "Thinking with 3 Pro" Muted greys and blues dominate the background. The button has a vivid yellow accent. " Select the Thinking with 3 Pro model" and tap "Create image" to begin. Panel 2: A cheerful person is depicted adding their selfie from the phone's camera. The user's face is drawn with soft outlines and warm pastel colors, while the phone and UI maintain the delicate water-ink aesthetic. Visible paper grain adds to the texture. Panel 3: The person thinks about what to create. In the background, different options are visualized to show what they're thinking, including — them as a plushie, them with a mohawk hairstyle, and a figurine. These options are clearly rendered behind the person in the same style as the rest of the comic. Panel 4: The person is shown adding a style prompt, with a speech bubble saying "Transform me into a watercolor painting". The text is integrated into the panel's watercolor-ink look, and the interaction feels natural and intuitive. Panel 5: The person is seen editing the image by simply typing into Gemini. The scene captures the ease of this interaction, with the final edited image, now in a watercolor style, appearing on the screen. The overall tone is friendly, instructional, and inspiring. It feels like a mini tutorial comic, all conveyed through the specified delicate water-ink illustration style. Make the aspect ratio 16:9.


Text / Vision
MLLM
Text
Diffusion
Vision
Text / Vision
Unified
Text / Vision

BAGEL: The Open-Source Unified Multimodal Model

RELEASED ON MAY 20, 2025

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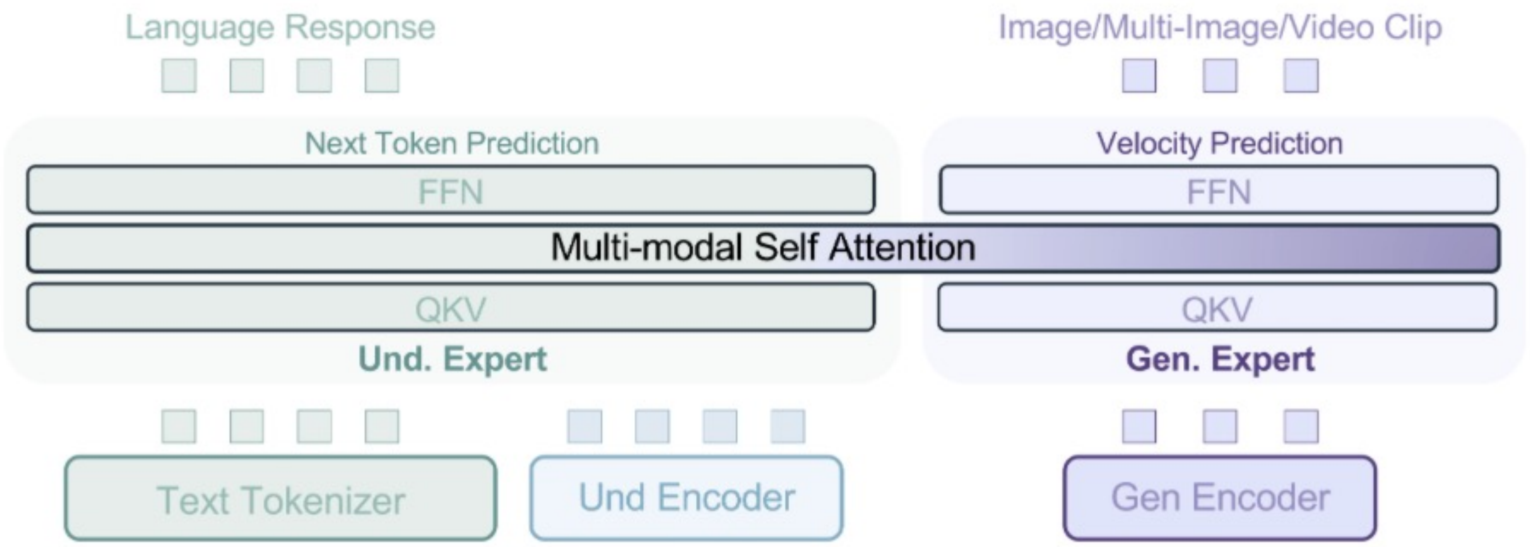
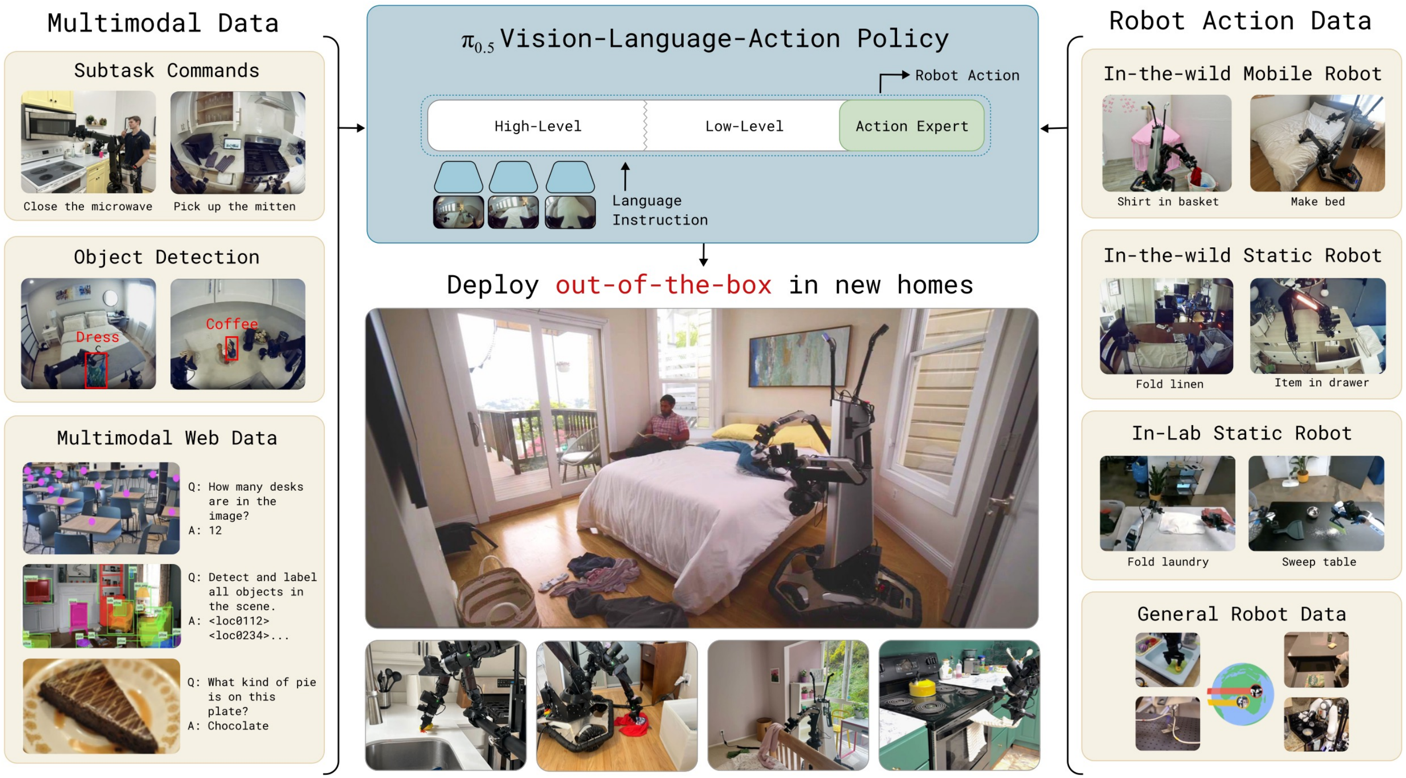
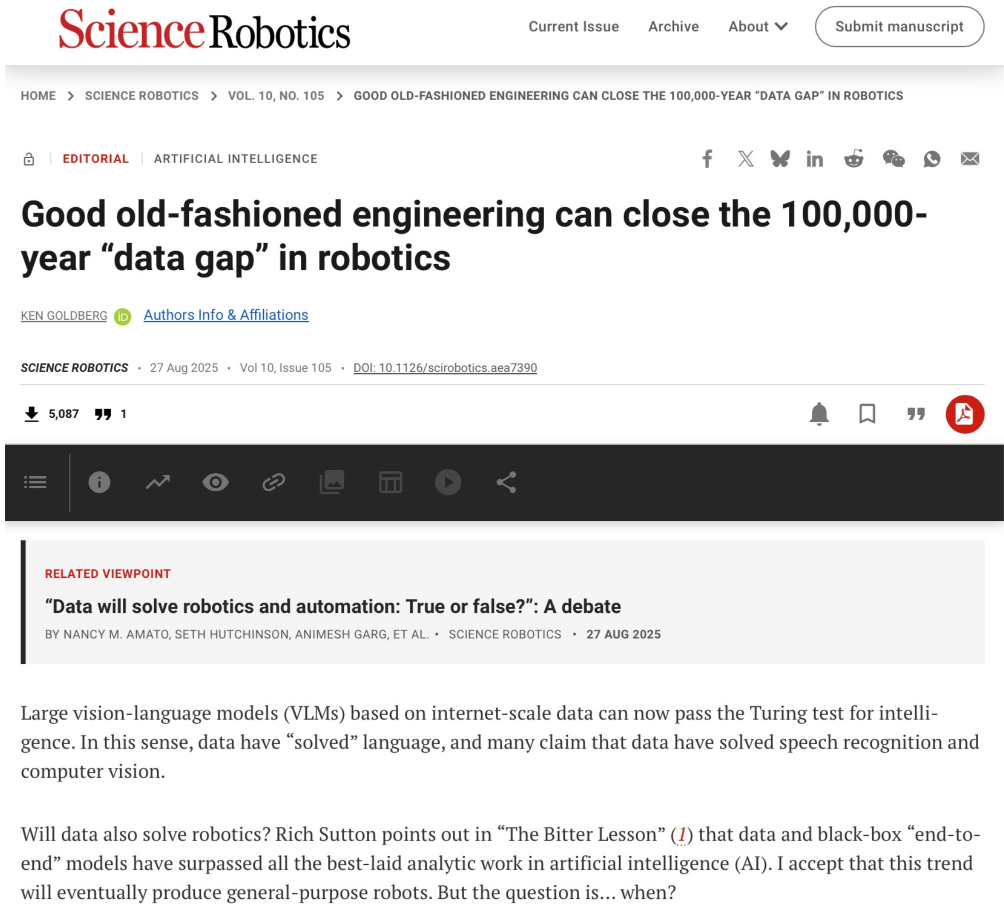


World Model





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EDITORIAL | ARTIFICIAL INTELLIGENCE

Good old-fashioned engineering can close the 100,000-year "data gap" in robotics

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SCIENCE ROBOTICS • 27 Aug 2025 • Vol 10, Issue 105 • DOI: 10.1126/scirobotics.aea7390

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RELATED VIEWPOINT

"Data will solve robotics and automation: True or false?": A debate

BY NANCY M. AMATO, SETH HUTCHINSON, ANIMESH GARG, ET AL. • SCIENCE ROBOTICS • 27 AUG 2025

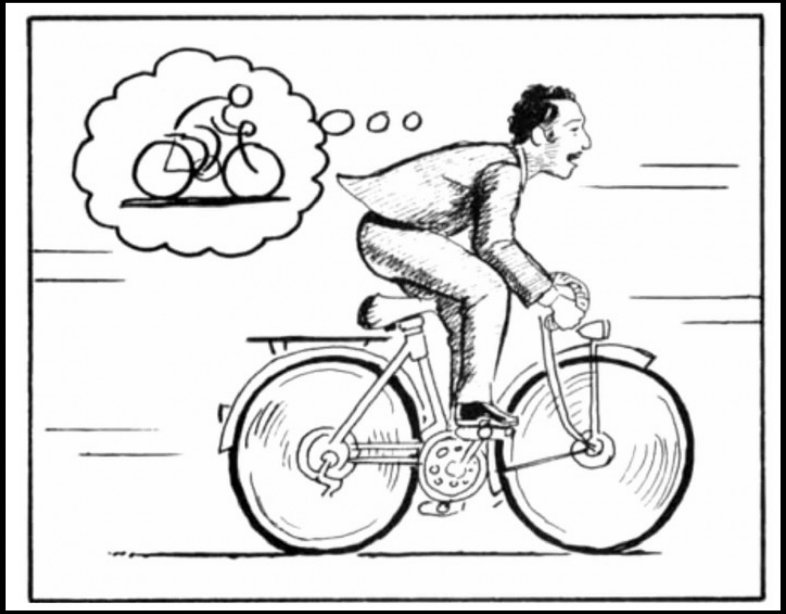
Large vision-language models (VLMs) based on internet-scale data can now pass the Turing test for intelligence. In this sense, data have "solved" language, and many claim that data have solved speech recognition and computer vision.

Will data also solve robotics? Rich Sutton points out in "The Bitter Lesson" (*1*) that data and black-box "end-to-end" models have surpassed all the best-laid analytic work in artificial intelligence (AI). I accept that this trend will eventually produce general-purpose robots. But the question is... when?

No basic animal learning process equivalent to supervised learning.

Animals learn from observing consequences and sequences of events, not from being shown the "right" action

Why humans learn so efficiently?



- Humans learn to act by observing the world.

- Humans build a "world model" of the real world in the brain:
  - Simulate situations;
  - Anticipate the outcomes of actions;
  - Learn skills through imagination.

World Model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s'


Siqiao Huang
@KnightNemo_
Not sure if now is the best time to do world model research, but it surely is good times for making world model memes🤣
Dudes showing up at the first conference for World Models...
JEPA
LLM
Google Map
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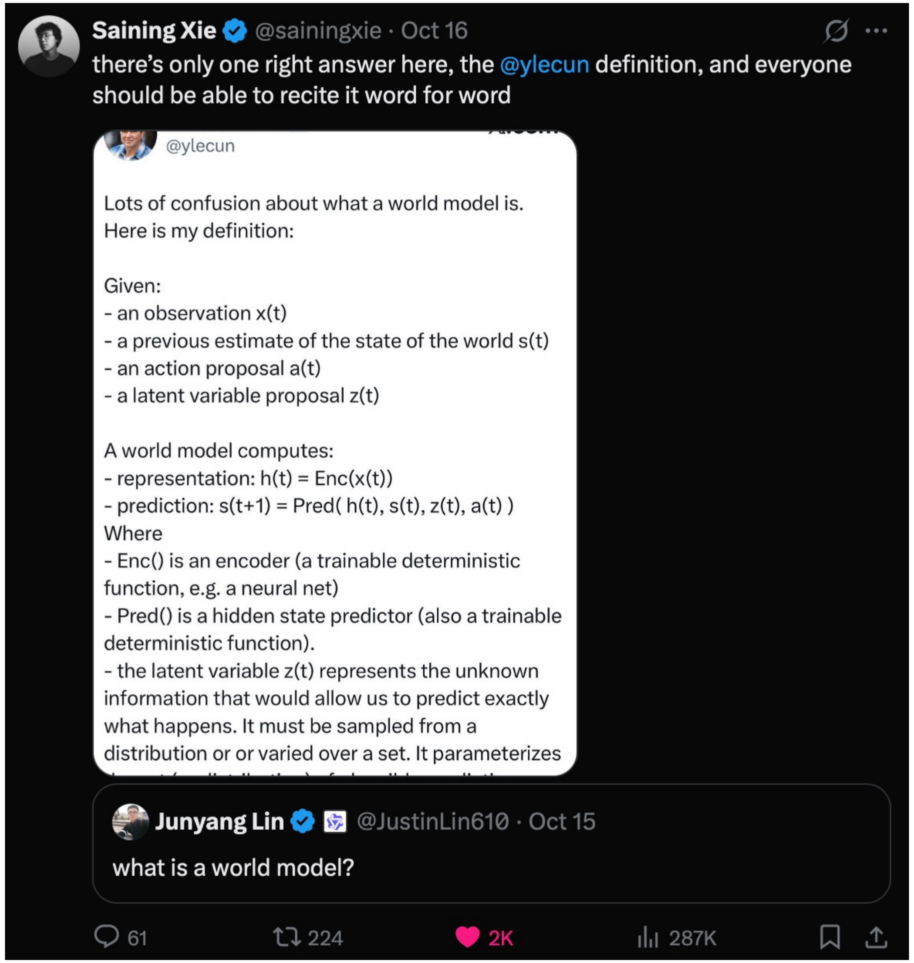
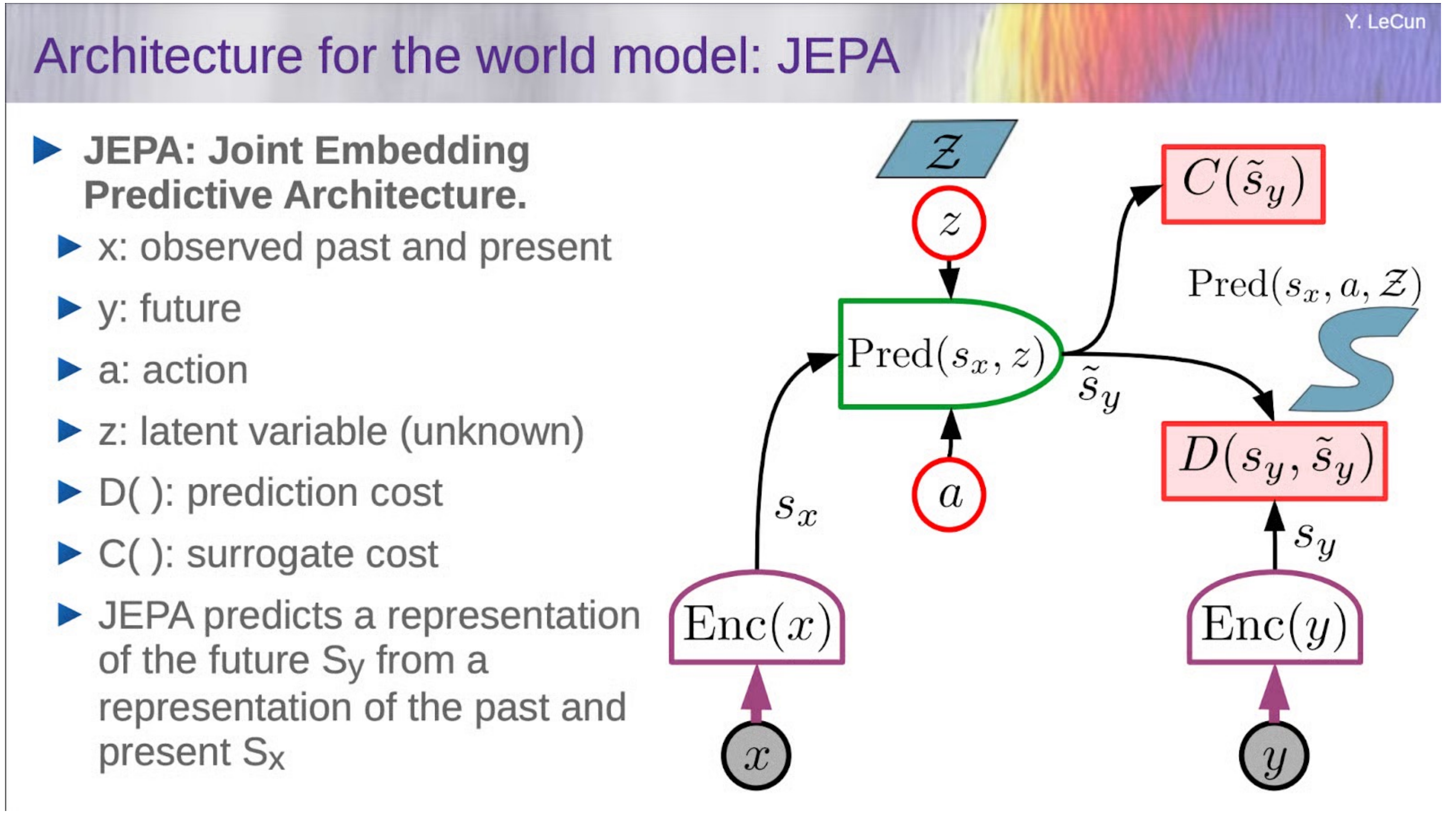
10:06 PM · Oct 18, 2025 · 8,108 Views

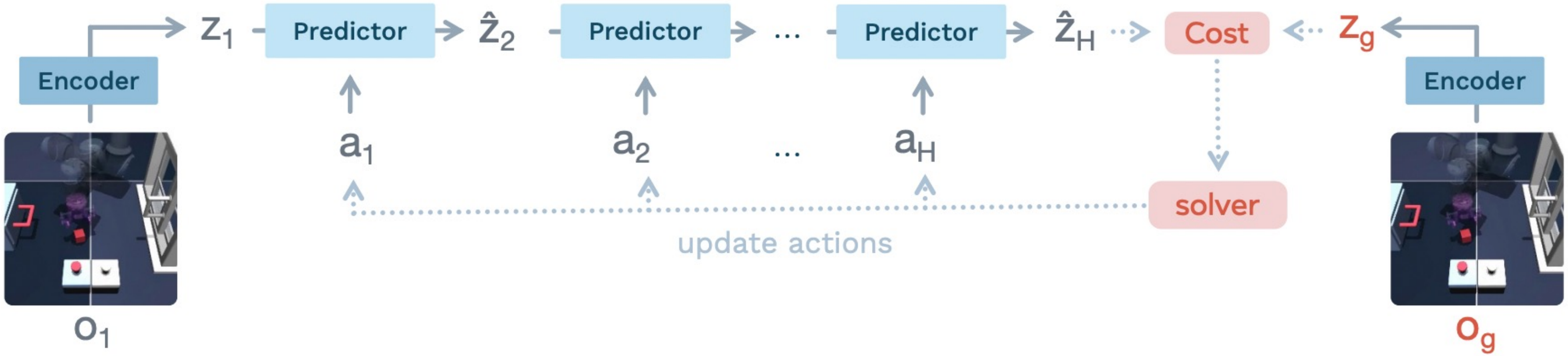

Saining Xie @sainingxie · Oct 16
there's only one right answer here, the @ylecun definition, and everyone should be able to recite it word for word
@ylecun
Lots of confusion about what a world model is.
Here is my definition:
Given:
- an observation x(t)
- a previous estimate of the state of the world s(t)
- an action proposal a(t)
- a latent variable proposal z(t)
A world model computes:
- representation: h(t) = Enc(x(t))
- prediction: s(t+1) = Pred(h(t), s(t), z(t), a(t))
Where
- Enc() is an encoder (a trainable deterministic function, e.g. a neural net)
- Pred() is a hidden state predictor (also a trainable deterministic function).
- the latent variable z(t) represents the unknown information that would allow us to predict exactly what happens. It must be sampled from a distribution or or varied over a set. It parameterizes
Junyang Lin @JustinLin610 · Oct 15
what is a world model?
61
224
2K
287K


Y. LeCun
Architecture for the world model: JEPA
JEPA: Joint Embedding Predictive Architecture.
x: observed past and present
y: future
a: action
z: latent variable (unknown)
D( ): prediction cost
C( ): surrogate cost
JEPA predicts a representation of the future Sy from a representation of the past and present Sx
$\mathcal{Z}$
z
$C(\tilde{s}_y)$
$\text{Pred}(s_x, a, \mathcal{Z})$
$\text{Pred}(s_x, z)$
$\tilde{s}_y$
$D(s_y, \tilde{s}_y)$
s_x
a
s_y
$\text{Enc}(x)$
$\text{Enc}(y)$
x
y

LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels

Lucas Maes*[1] Quentin Le Lidec*[2] Damien Scieur[1,3] Yann LeCun[2] Randall Balestriero[4]

[1]Mila & Université de Montréal [2]New York University [3]Samsung SAIL [4]Brown University



World Model

Causal

Interactive

Real-time

Physical

Causal

Interactive

Real-time

Physical

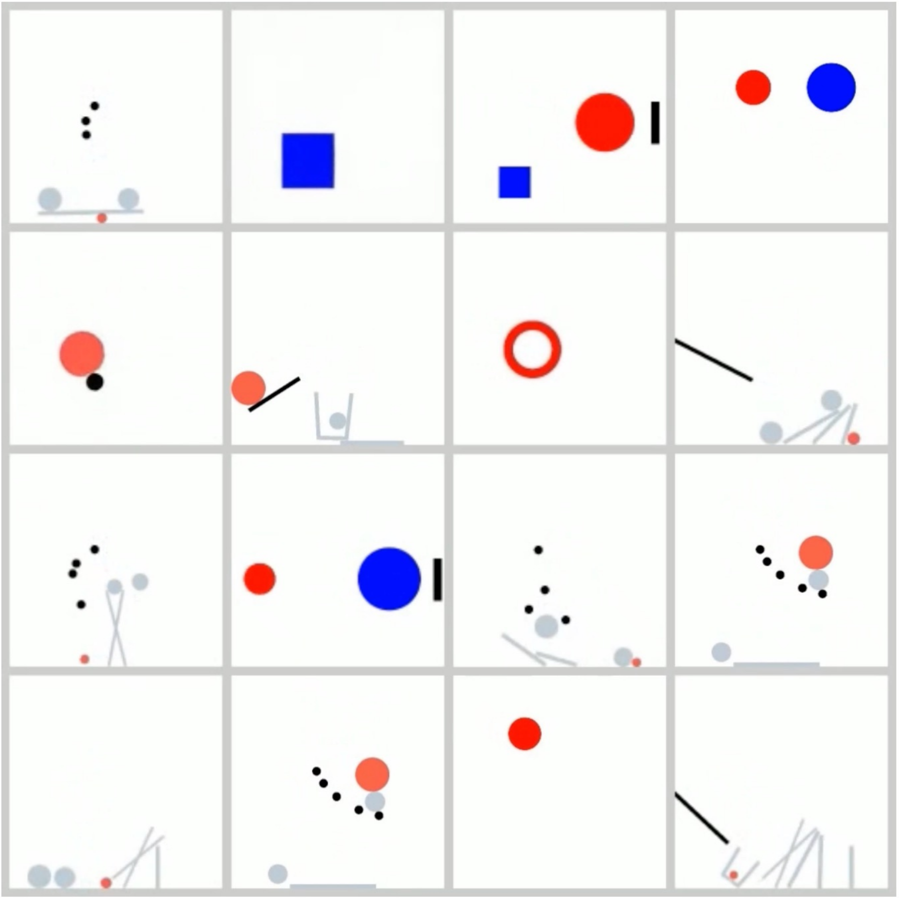
How Far is Video Generation from World Model?

— *A Physical Law Perspective*

We conduct a systematic study to investigate whether video generation is able to learn physical laws from videos, leveraging ***data and model scaling***.

- **Generalization Scenarios**: We consider three scenarios: in-distribution, out-of-distribution and combinatorial generalization.
- **Tasks**: We consider physical events governed by one or more classical mechanics laws, e.g., *law of inertia*, *Newton's second law* and *the conservation of energy*.
- **Data**: We develop a 2D simulator with simple geometric shapes (eliminating textures), ensuring unlimited supply of data for scaling.
- **Model**: We use standard video generation model and focus on scaling-up.
- **Key Messages**: We first summarize the ***generalization observations*** of video models, then analyze their underlying ***generalization mechanism*** and ***priority***.
- **Community**: Click to view the lively discussions within the community.

arXiv | pdf | Code | Data

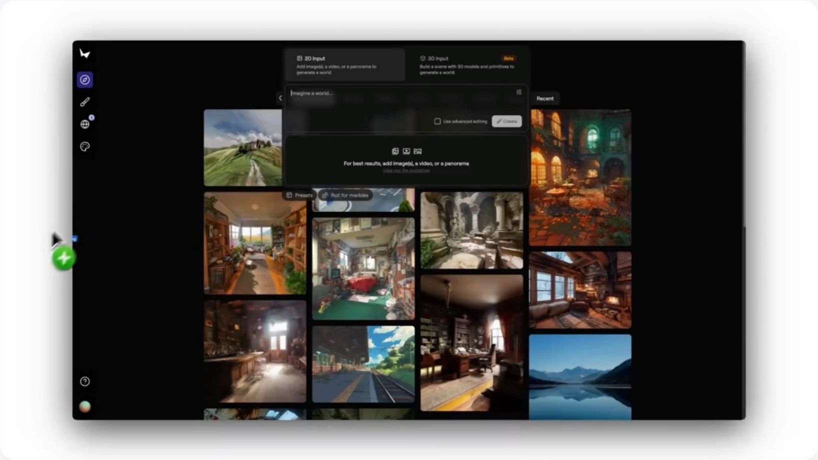


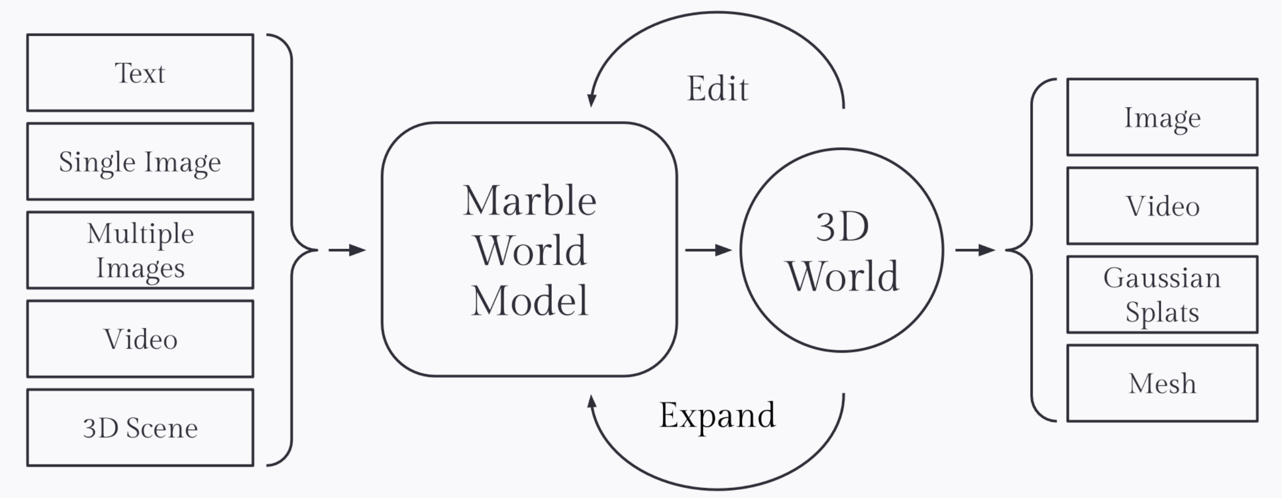
From pixels to worlds

Marble, our first product, generates spatially consistent, high-fidelity, and persistent 3D worlds that you can move through, edit, and inhabit.

- **Multimodal inputs**
  Create detailed 3D worlds from text, images, videos, or 360 panoramas.
- Create from 3D layouts
- Interactive editing
- Expand and combine worlds
- Versatile outputs

Start creating with Marble





Grok AI falsely accuses NBA star of vandalism spree

In an April 2024 post on X, Grok, the AI chatbot from Elon Musk's xAI, falsely accused NBA star Klay Thompson of throwing bricks through windows of multiple houses in Sacramento, Ca.

NYC AI chatbot encourages business owners to break the law

In March 2024, *The Markup* reported that Microsoft-powered chatbot MyCity was giving entrepreneurs incorrect information that would lead to them break the law.

in October 2024, MyCity was intended to help provide New Yorkers with
on on starting and operating businesses in the city, as well as housing
d worker rights. The only problem was *The Markup* found MyCity falsely
ousiness owners could take a cut of their workers' tips, fire workers
of sexual harassment, and serve food that had been nibbled by
o claimed landlords could discriminate based on source of income.

3. Fake citations from ChatGPT used by an attorney

In May, the news broke that Steven Schwartz will be charged in a New York court for using fake citations from OpenAI's ChatGPT in legal research for a case he was handling.

iTutor Group's recruiting AI rejects applicants due to age

In August 2023, tutoring company iTutor Gro
a suit brought by the US Equal Employment
The federal agency said the company, which
students in China, used AI-powered recruitin
rejected female applicants ages 55 and older
older.

5. Uber Eats began using AI-generated food images

Take care when ordering takeouts. Uber Eats is using AI for pictures of food when none are provided by the restaurant but the tech couldn't tell the difference when the same term — "medium pie" — was used to describe a pizza at an Italian restaurant and a sweet dessert. In the same example, it invented a brand of ranch dressing called "Lelnach" and showed a bottle of that in the picture.

Domain Taxonomy of AI Risks

The Domain Taxonomy of AI Risks classifies risks from AI into seven domains and 23 subdomains. You can explore the taxonomy (to four levels of depth) in the interactive figure below. Read **our preprint** for more detail.

1. Discrimination & Toxicity
2. Privacy & security
3. Misinformation
4. Malicious actors & misuse
5. Human-computer interaction
6. Socioeconomic & environmental harms
7. AI system safety, failures and limitations